SHINRYO VIETNAM

SHINRYO VIETNAM CORPORATION

HO CHI MINH HEAD OFFICE
3/F., Saigon Prime Office Building
107-109-111 Nguyen Dinh Chieu St.,
Ward 6, Dist. 3, HCMC Vietnam
Tel No. : 0084 (28) 39300847 / 8, 39302184 / 5
Fax No. : 0084 (28) 39302181
Email : svn@shinryovn.com

HANOI BRANCH OFFICE
4/F., Lac Hong Office Building,
85 Le Van Luong St., Nhan Chinh Ward
Thanh Xuan Dist., Hanoi, Vietnam
Tel No. : 0084 (24) 37660891
Fax No. : 0084 (24) 37660901

PURCHASE ORDER

MESSRS.
THUAN PHONG INNOVATIVE TELECOMMUNICATION CO., LTD
No 10, 3/2 Street, Ward 12, District 10, HCMC

Attn: Mr. Huynh Thanh Hai - Tel : 028 3862 1119

NO. **6001403200-038** DATE 17 Jun.2020

Please supply the following item(s) and deliver to : JOTUN VIETNAM GREENFIELD PROJECT – MEP PACKAGE

Quotation Ref.	Delivery Date	Terms of Payment	For query, please contact	Project Name
TBM-JOTUN Date: 13 Jun.2020	14 Aug.2020	T/T	Mr. Luong	JOTUN VIETNAM GREENFIELD PROJECT - MEP PACKAGE

Our Budget Code	Description	Unit	Quantity	Net Unit Price	Net Amount
178-0001	TELECOMMUNICATION SYSTEM Attached Telecommunication system specification and schedule of unit rate (Total: 054 pages) 1. Delivery Address: Jotun Vietnam Greenfield Project - MEP Package: - Street No.24, Hiep Phuoc Industrial Park, Nha Be Dist., HCMC - Contact person: Mr. Trung (HP: 0909-634-609) 2. Term of payment: According to SVN's regular payment schedule	Lot	1	VND VAT 10% TOTAL	2,693,990,000 269,399,000 2,963,389,000

ORIGINAL

IMPORTANT
1. This Purchase Order is issued subject to the Terms and Conditions overleaf.
2. Please acknowledge this Order by signing and returning the duplicate hereof immediately.
3. To avoid delay in payment, please submit your invoice in DUPLICATE, quoting our PURCHASE ORDER NUMBER.

CÔNG TY TNHH SHINRYO VIỆT NAM
M.S.D.N:0304968829-C.T.T.N.H.H
QUẬN 3 - T.P HỒ CHÍ MINH

For SHINRYO VIETNAM CORPORATION

Authorized Signature

Maker	Checked by

Total Value of This Order Not to Exceed: **VND 2,963,389,000**

Amount Say : VND Two billion nine hundred sixty three million three hundred eighty nine thousand only

Sep - 2015

SVN-F-03-04 Rev.0

TERMS AND CONDITIONS OF PURCHASE

1. ACCEPTANCE

Agreement by Seller to furnish the materials or services hereby ordered or its furnishing such materials or services in whole or in part or the commencement of work by the Seller with reference thereto shall constitute acceptance by Seller of this order subject to these terms and conditions. In the event that the order does not state price or delivery. Buyer will not be bound to any prices or delivery to which it has not specifically agreed in writing. Any terms and conditions of purchase herein contained shall be void and of no effect, unless specifically agreed to by Buyer. Modifications hereof or additions hereto, to be effective, must be rnade in writing and be signed by Buyer. These terms and conditions together with such modifications and wilh such data relating to price and delivery as are accepted in writing by Buyer constitute the entire agreement between the parties. The rights of both parties hereunder shall be in addition to any of its rights and remedies at law. Failure of Buyer to enforce its rights shall not constitute a waiver of such rights or of any other rights.

2. PRICES

The acceptance of this order constitutes a warranty that the prices to be charged for articles or services ordered herein are not in excess of prices charged to other customers for similar quantities and delivery requirements

3. TIME OF DELIVERY

It is understood and agreed that time is of the essence of this order.

4. PACKING AND DELIVERY

Unless otherwise specified, price is to cover net weight of material ordered hereunder and no charges will be 'allowed for boxing, crating, carting, or storage. Without the purchaser's prior written consent deliveries aqainst this order shall not be made in whole or in part prior to the date or dates shown hereon nor shall they exceed the quantities specified.

5. QUANTITIES OVER-SUPPLIED

Buyer will pay only for maximum quantities ordered Ouantities over-supplied will be at Seller's risk. Return of excessive quantities will be undertaken by Seller at its expense.

6. INSPECTION

The articles and all parts, material and workmanship entering into the performance of this order shall be subject to inspection, test and count by Buyer at all times and places whether during or after manufacture. If anv of the articles shall be detective in material or workmanship or otherwise not in conformity with the requirements or this order. Buyer, in addition to its other rights, may reject the same for full credit or may rework same at Seller's expense or require prompt correction or replacement thereof at Seller's expense, including transportation charges.

7. WARRANTY

Seller warrants that all material ordered hereunder will conform in all respects with the specifications, drawings, sample or other description furnished or specified by the Buyer, and will be merchantable and free from any defects in material and workmanship, and Seller further warrants that all material purchased hereunder that is manufactured in accordance with the Seller's specifications shall be fit and sufficient for the purposes for which it was designed Seller agrees that the foregoing warranty shall survive acceptance of and payment for the material and shall save Buyer harmless from any loss, damage of expense, whatsoever, including attorney's fees, that Buyer may incur as a result of any breach of such warranties.

8. INVOICE AND PAYMENT

Individual invoices in duplicate, quoting this Purchase Order number must be issued for each delivery of materials or stage of services rendered. Payment of invoice shall not constitute acceptance of supplies or services and shall be subject to adjustment for rejection, errors, shortage, defects in supplies or services or other failure of Seller to meet the requirements of this Order.

9. CHANGES

The Buyer may at any time, by a written order, make changes, within the general scope of this order in any one or more of the following.

(a) applicable drawings designs, or specifications;
(b) method of delivery or packing.
(c) place of delivery

If any such change causes an increase or decrease in the cost of, or the time required for performance of this order, an equitable adjustment shall be made in the order price or delivery schedule or both, and the order shall be modified in writing accordingly. Any claim by the Seller for adjustment hereunder must be asserted within 20 days from the date of receipt by the Seller of the notification of change provided, however, that such period may be extended upon the written approval of the Buyer. However, nothing in this caluse shall excuse the Seller from proceeding with the order as changed or modified

10. TERMINATION FOR DEFAULT

(a) The Buyer may terminate all or any part of this order without liability to the Seller, by written notice of default, if Seller fails to perform its obligations under this order as specified or so fails to make progress as to endanger performance under this order and in accordance with its terms. The Buyer is the sole judge under such circumstances.

(B) In the event of Seller's default or potential inability to perform this order, Seller agrees upon demand by Buyer to deliver to Buyer the raw materials and work in process acquired in order to perform under this order, and Buyer may then complete the work deducting the cost of such completion from the price, or in the alternative pay to Seller the cost of such raw materials and work in process.

11. BUYER'S PROTECTION IN CONNECTION WITH WORK DONE AT ITS PLANT

The Seller shall take such steps as may be reasonably necessary to prevent personal injury or property damage during any work hereunder that may be performed by any employees or aqents or subconstructors of the Seller at Buyer's plant and the Seller shall indemnify and hold harmless the Buyer from and against all loss, liability, and damages arising from or caused solely by any act of omission of such agents, employees of subconstructors of the Seller and Seller shall maintain such insurance against public liability and property damage, and such Employee's Liability and Compensation Insurance as will protect the Buyer against the aforementioned risks and against any claims under any Workmen's Compensation and Occupational Disease Acts.

12. GRATUITIES

Seller warrants that it has not offered of given and will not offer or give to any employee, agent or representative of Buyer any gratuity with a view toward securing any business from Buyer or influencing such person with respect to the terms, conditions or performance of any contract with or order from Buyer. Any breach of this warranty shall be a material breach of each and every contract between Buyer and Seller.

13. NON-DISCLOSURE OF CONFIDENTIAL MATTER

Materials purchased hereunder with the Buyer's specifications of drawings shall not be quoted for sale to others without the Buyer's written authorization. Such specifications, drawings, samples or other data furnished by the Buyer shall be treated as confidential information by the Seller shall remain Buyer's property and shall be returned to it on request.

14. ASSIGNMENT

No right or obligation under this order (including the right to receive moneys due and to become due hereunder), shall be assigned by Seller without the prior written consent of Buyer, and any purported assignment without such consent shall be void.

15. BUYER-FURNISHED PROPERTY

All tools or other materials furnished by the Buyer for use in the performance of this order shall remain the property of fhe Buyer, shall be used by the Seller in the performance of this order only in accordance with the requirements of the order relating to such use, and shall be returned to the Buyer when requested upon the completion or termination of the order to the extent not previously delivered to the Buyer.

16. SPECIAL TOOLING

If all of the costs of special tooling used in the performance of this order have been charged to this order or to this order and other orders placed by the Buyer, title to such special tooling shall vest in the Buyer, at the option of the Buyer Such tooling is to be used only in the performance of such Purchase Order unless otherwise approved by the Buyer. The Seller agrees that it will follow normal industrial practice in the identification and maintenance of property control records on all such tooling, and will make such records available for inspection by the Buyer at all reasonable times. After the termination of completion of such order(s) and upon the request of the Buyer, the Seller shall furnish a list of such tooling in the form requested and shall make such tooling available for disposition by the Buyer.

17. GOVERNING LAW

This Purchase Order shall be governed by the laws of Japan and constitutes the entire agreement between Buyer and Seller.

SHINRYO VIETNAM

SHINRYO VIETNAM CORPORATION

HO CHI MINH HEAD OFFICE
3/F., Saigon Prime Office Building
107-109-111 Nguyen Dinh Chieu St.
Dist. 3, HCMC, Vietnam
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HANOI BRANCH OFFICE
4F., Lac Hong Office Building
85 Le Van Luong St., Nhan Chinh Ward
Thanh Xuan Dist., Hanoi, Vietnam
Tel No. : 0084 (24) 37660891
Fax No.: 0084 (24) 37660901

Date:	**17 Jun. 2020**
Ref.:	**SVN-6001403200-038**
To:	**THUAN PHONG INNOVATIVE TELECOMMUNICATION CO., LTD**
Attention:	**Mr. Huynh Thanh Hai**
From:	**Mr. Kawano/ Mr. Thang/ Mr. Luong**
Project:	**JOTUN VIETNAM GREENFIELD PROJECT – MEP PACKAGE**
Re:	**PURCHASE ORDER FOR "Telecommunication system "**
	No.: 6001403200-038

Dear Sir,

Further to our discussion and your Final Quotation No.TBM-JOTUN dated 13 Jun. 2020, we hereby would like to issue an official "Purchase Order" as attached original and yellow duplicate.

Please sign and return the YELLOW DUPLICATE immediately (within 05 days) as your acknowledgement of our order to below address:

SHINRYO VIETNAM CORPORATION
3/F., Saigon Prime Office Building, 107-109-111 Nguyen Dinh Chieu Street
District 3, Ho Chi Minh City, Vietnam
Attention: Ms. Nu
Tel: +84 (028) 39300847 / 39300848 Fax: +84 (028) 39302181

*** PLEASE WRITE DOWN OUR P.O NO. ONTO YOUR DELIVERY NOTE AS WELL AS CLAIM INVOICE OR VAT INVOICE ALWAYS.**

Should you have any queries, please feel free to contact us.

Thank you very much for your cooperation.

Yours faithfully,

For on and behalf of

SHINRYO VIETNAM CORPORATION

KIICHI KAWANO
Director

Details for issue VAT Invoice (if any):
- **CÔNG TY TNHH SHINRYO VIỆT NAM**
- **Cao ốc văn phòng Sài Gòn Prime, 107-109-111 Nguyễn Đình Chiểu, P.6, Q.3, HCM**
- **Tax code/MST: 0304968829**

Encl. as state

SHINRYO VIETNAM CORPORATION
CÔNG TY TNHH SHINRYO VIỆT NAM

Project: JOTUN VIETNAM GREENFIELD PROJECT – MEP PACKAGE
Package: TELECOMMUNICATION SYSTEM
Vendor: THUAN PHONG INNOVATIVE TELECOMMUNICATION CO., LTD
Purchase Order No.: 6001403200-038 Date: 17 Jun.2020
Attachment

CÔNG TY TNHH SHINRYO VIỆT NAM M.S.D.N: 0304908829-C.T.C ★ QUẬN 3 - T.P HỒ CHÍ MINH

1. Telecommunication system Specifications and Schedule of Unit Rate:

Technical submittal referenced VD156-SVN-JOTUN/TVC-MA-0071 dated 12 May.2020

Item	Description	Quantity	Unit	Unit price (VND)	Total Amount (VND)	Country of Origin
1	Switch Aruba 2530 48G PoE + Model: J9772A	9	pc	37,000,000	333,000,000	Asia
2	Switch HPE 1920S 24G 2SFP PoE + 370W Model: JL385A	8	pc	17,800,000	142,400,000	Asia
3	Switch Aruba 1G SFP LC SX 500m MMF transceiver Model: J4858D	34	pc	3,980,000	135,320,000	Asia
4	Access point Meraki MR42 cloud managed AP Model: MR42-HW	18	pc	14,500,000	261,000,000	Asia
5	Access point Meraki MR74 cloud managed AP Model: MR74-HW	51	pc	19,890,000	1,014,390,000	Asia
6	Access point Meraki Dual-band omni antennas Model: MA-ANT-20	102	pc	2,880,000	293,760,000	Asia
7	Access point Meraki MR Enterprise license, 5YR Model: LIC-ENT-5YR	69	pc	6,480,000	447,120,000	Asia
8	Switch programming	1	lot	67,000,000	67,000,000	
Final Purchase Order Amount before VAT (VND):					**2,693,990,000**	

2. Terms and Conditions:

a. Term of Payment:

- 90% of contract amount after delivery goods, shipping document and according to SVN's payment schedule.
- Shipping document is required as below:

  + Certificate of origin issued by authorities for import products.

  + Certificate of Quantity and Quality

  + Detailed packing list.
- 10% of contract amount after T&C and SVN receives the warranty certificate in 24 months from 30 April 2021 as enclosed and according to SVN's payment schedule.

- All tax invoices and delivery orders shall be submitted to us and invoices shall reach our office on/before last certified day of each month.
- Payment shall be made by bank transfer on/after the day of 7 and 22 of every month.

b. Warranty Period:

Warranty period shall be (24) twenty-four months from the hand-over date. The expected hand-over date for whole of the Works is 30 April 2021, subject to the issuance of the Certificate of Completion.

c. Penalty for Delay Delivery:

In case of delayed shipment and subsequent late delivery of the aforesaid equipment/materials, or delay in completion of the works, all cost incurred by Shinryo shall be to your account.

CÔNG TY TNHH TTCL VIỆT NAM
TTCL VIETNAM CORPORATION LIMITED
Enterprise Code: 0300790729
11th Floor, Centre Point Building, 106 Nguyen Van Troi Street,
Ward 8, Phu Nhuan District, Ho Chi Minh City, Vietnam.
Tel.: +(84-8) 39977118 Fax.: +(84-8) 39977086

Company : SHINRYO VIETNAM CORP.
Add : *107-109-111 Nguyen Dinh Chieu Street*
Tel : *+84(08) 3930-0847/48*
Fax : *+84(08) 3930-2181*

JOTUN VIETNAM – GREEN FIELD PROJECT
Transmittal. No. : VD156-SVN-JOTUN/TVC-T-0313
Date : 12/May/2020
Jotun's Job No.: 2017-2340
TVC Job No. : VD-156

Attn : Mr. Duoc_Project/Construction Manager
Subject : MA – Telecommunication system Rev.01

Dear Sirs,

We are enclosing herewith one (1) original of document & drawings indicated below for

☐ Review ☐ Information ☒ Approval ☐ Construction
☐ Final ☐ As-Built ☐ Reference

No.	Doc. / Dwg. No.	Title	Rev. No.	Remarks
See the attached file (1 hard copy):				
1	VD156-SVN-JOTUN/TVC-MA-0071	MA – Telecommunication system	01	

Best Regards,

Mr. Nguyen Dang Khoa
Project/Construction Manager

Please Return one (1) copy of this form with your signature, for your acknowledge of receipt.

Signature : N.T.G. My

Date : May/12/20 . 5:10 pm

Form 1

VENDOR PRINT CONTROL SHEET (VPCS)

RECEIVED DATE :
ISSUED DATE :

Vendor : *SHINRYO VIETNAM CORP.*

Jotun/TVC Job No. : 2017-2340/VD-156
P/O No. :
Item No. or Service :
Vendor Transmittal No. : VD156-SVN-JOTUN/TVC-T-0373
Vendor Issue Date : 12-May-20

JOTUN VIETNAM - GREENFIELD PROJECT

OWNER	CONSULTANT (TVC)	
PM	CM	LEADER
[signature]	[signature]	[signature]

SUBJECT : RESPONSE ON YOUR TRANSMITTED DRAWING AND/OR DOCUMENTS (VPCS)

Dear Sirs, The followings are our response on your transmitted drawings and/or documents as listed below.
If you find any our comments, please take them into your related drawings and/or documents.

MARK	PROJECT'S DOC./DWG. NO.	REV.NO.	DOCUMENT / DRAWING TITLE	SUBMISSION FOR	TVC'S RESPONSE AP	AN	NA	RE SUB-MISSION REQUESTED DATE
1	VD156-SVN-JOTUN/TVC-MA-0071	1	MA– Telecommunication system	FA	√			

The APPROVED stated above does not relieve the Vendor or his responsibility to meet with all requests of the purchase order or and the Mutual Agreement in writing.

Note 1 : AP:Approved (No comments and released for fabrication), AN: Approvved as Note and updated in Final submission, NA : Not Approved and Re-submit

Note 2 : Submission For; FA:For Approval, FI:For Information, FF:For Final , AB:As-Built

Note (from TVC to Vendor) :

Note (for TVC Internal) :

For Purchaser's use

Distribution / Serial No.		Vendor	Owner	Others	TVC Project	Site	PR	ME	MA	PP	EE	IN	CV	Proc	QC
	VPCS														
C															
O															
P															
Y															

LEGEND

Form 3

CONTRACTOR DOC. NO: VD156-SVN-JOTUN/TVC-MA-0071

TOTAL 402 SHEET

SERVICE : MEP PACKAGE

DOCUMENT TITLE : MA - TELECOMMUNICATION SYSTEM REV.01

JOTUN VIETNAM - GREENFIELD PROJECT
CLIENT: **JOTUN PAINTS VIETNAM CO., LTD** **Jotun Job No. 2017-2340**
CONSULTANT: **TTCL VIETNAM CORPORATION LIMITED** **TVC's Job no. VD156**
CONTRACTOR: **SHINRYO VIETNAM CORP.**
Area No :
Item No.:
Project Doc. No. VD156-SVN-JOTUN/TVC-MA-0071 — Rev. 01

✓	APPROVED
	APPROVED AS NOTED & RESUBMIT
	NOT ACCEPTED & RESUBMIT
	REVIEW WITH COMMENT
	REVIEW & NOT RETURN

Acceptance in any of these categories in the way relieves the contractor/supplier of his responsibilities for all requirements in accordance with the contract

DATE: 14-05-2020

CM	Checker	PM	Checker
EPCM		OWNER	

FOR APPOROVAL

01	12-May-20	FOR APPROVAL	N.C.Hoat	L.Q.Trung	N.D.Khoa	
0	4-Apr-20	FOR APPROVAL	N.C.Hoat	L.Q.Trung	N.D.Khoa	
REV.	DATE	DESCRIPTION	PREP' D	CHECKED	APPROVED	AUTH' D

MATERIAL SUBMISSION

Subject: TELECOMMUNICATION SYSTEM

Table of Content:

1. Equipment Schedule

NO	DESCRIPTION	SPECIFICATION (Tender Requirement)	SPECIFICATION (Tender Submission)	SPECIFICATION (For Approval)	APPLICATION	BRAND	ORIGIN	REMARKS
1	PoE switch 48 port	Power supply 230VAC, 50Hz Operating temperature: 0° to 40°C Operating humidity: 10% to 90%	Power supply 230VAC, 50Hz Operating temperature: 0° to 45°C Operating humidity: 15% to 95% Model: J9772A	Power supply 230VAC, 50Hz Operating temperature: 0° to 45°C Operating humidity: 15% to 95% Model: J9772A	Telecommunication System	HP/Aruba (comply)	ASIA	
2	PoE switch 24 port	Power supply 230VAC, 50Hz Operating temperature: 0° to 40°C Operating humidity: 10% to 90%	Power supply 230VAC, 50Hz Operating temperature: 0° to 40°C Operating humidity: 15% to 95% Model: JL385A	Power supply 230VAC, 50Hz Operating temperature: 0° to 40°C Operating humidity: 15% to 95% Model: JL385A	Telecommunication System	HP/Aruba (comply)	ASIA	
3	Wireless Access Point Meraki MR42	Meraki MR33 Cloud Managed AP Meraki Dual-band Onmi Antennas Meraki MR Enterprise License, 5YR	Meraki MR33 Cloud Managed AP Meraki Dual-band Onmi Antennas Meraki MR Enterprise License, 5YR	Meraki MR42 Cloud Managed AP Meraki Dual-band Onmi Antennas Meraki MR Enterprise License, 5YR	Telecommunication System	Cisco-Meraki (Comply)	ASIA	
4	Wireless Access Point Meraki MR74	Meraki Dual-band Omni Antennas (4 poles antennas) Meraki MR74 Cloud Managed AP Meraki MR Enterprise License , 5YR	Meraki Dual-band Omni Antennas (4 poles antennas) Meraki MR74 Cloud Managed AP Meraki MR Enterprise License , 5YR	Meraki Dual-band Omni Antennas (4 poles antennas) Meraki MR74 Cloud Managed AP Meraki MR Enterprise License , 5YR	Telecommunication System	Cisco-Meraki (Comply)	ASIA	

2. Catalogue

2.1. PoE switch 48 port

DATA SHEET

ARUBA 2530 SWITCH SERIES

Switch POE 48Port - J9772A

PRODUCT OVERVIEW

The Aruba 2530 Switch Series provides cost-effective, reliable and secure access layer connectivity for enterprises, branch offices and small and midsize businesses.

These fully managed switches deliver Layer 2 capabilities with enhanced access security, traffic prioritization, sFlow, and IPv6 host support. Right size deployment is available with a range of Gigabit and Fast Ethernet models including compact and fanless models which are ideal for use in quiet work spaces. PoE+ models deliver up 370W to power access points, IP phones and cameras.

The Aruba 2530 Switch Series is easy to deploy, use and manage using Aruba AirWave or Aruba Central. Aruba ClearPass offers Network Access Control and external captive portal support. The switches include a Limited Lifetime Warranty.

KEY FEATURES

- Cost-effective, reliable and secure Aruba Layer 2 switch series
- Flexible Management via Aruba AirWave, Aruba Central, and Aruba ClearPass Policy Manager
- Right size deployment with choice of 8, 24 and 48 port Gigabit and Fast Ethernet models
- Up to 370W PoE+ to power IoT, APs and cameras
- REST API support
- Simple deployment with Zero Touch Provisioning

ENHANCED FEATURES

Wired and Wireless Support

- Switch auto-configuration automatically configures switch for different settings such as VLAN, CoS, PoE max power, and PoE priority when an Aruba access point is detected
- Local User Role defines a set of switch-based policies in areas such as security, authentication, and QoS. A User Role can be assigned to a group of users or devices, using local switch configuration (YA releases only)

Quality of Service (QoS)

- Traffic prioritization (IEEE 802.1p) for real-time traffic classification. Support for eight priority levels mapped to either two or four queues, and uses weighted deficit round robin (WDRR) or strict priority
- Simplified quality of service (QoS) configuration
  - Port-based traffic prioritization by specifying a port and priority level
  - VLAN-based traffic prioritization by specifying a VLAN and priority level
- Class of Service (CoS) sets the IEEE 802.1p priority tag based on IP address, IP Type of Service (ToS), Layer 3 protocol, TCP/UDP port number, source port, and DiffServ
- Rate limiting establishes per-port ingress-enforced maximums for all traffic or for broadcast, multicast, or unknown destination traffic
- Layer 4 prioritization enables priorities based on TCP/UDP port numbers
- Flow control delivers reliable communication during full-duplex operation

Simplified Configuration and Management

- Aruba Central cloud-based management platform offers a simple, secure and cost effective way to manage switches. Complies with RFC 7030 for encryption key enrollment
- Zero-Touch-Provisioning (ZTP) simplifies installation of the switch infrastructure using DHCP-based process with AirWave
- Flexible management - Supports both cloud-based Central and on-premise AirWave without ripping and replacing switching infrastructure
- Choice of management interfaces
  - HTML-based easy-to-use Web GUI allows configuration of the switch from any Web browser
  - Robust CLI provides advanced configuration and diagnostics
  - Simple network management protocol (SNMPv1/v2c/v3) allows the switch to be managed with a variety of third party network management applications

- Provides single IP address management for up to 16 switches individually
- sFlow® (RFC 3176) delivers wire-speed traffic accounting and monitoring, configured by SNMP and CLI with three terminal encrypted receivers
- IEEE 802.1AB Link-Layer Discovery Protocol (LLDP) automates device discovery protocol for easy mapping by network management applications
- Provides local and remote logging of events via SNMP (v2c and v3) and syslog; provides log throttling and log filtering to reduce the number of log events generated
- Port mirroring allows traffic to be mirrored on any port or a network analyzer to assist with diagnostics or detecting network attacks
- Remote monitoring (RMON) provides advanced monitoring and reporting capabilities for statistics, history, alarms, and events
- Friendly port names allows assignment of descriptive names to ports
- Dual flash images provides independent primary and secondary operating system files for backup while upgrading
- Multiple configuration files are easily stored with a flash image
- Front-panel LEDs
  - Locator LEDs allows users to set the locator LED on a specific switch to turn on, blink, or turn off; and simplifies troubleshooting by making it easy to locate a particular switch within a rack of similar switches
  - Per-port LEDs provides an at-a-glance view of the status, activity, speed, and full-duplex operation
  - Power and fault LEDs display issues, if any

Connectivity

- Compact and fanless 8-port models offer quiet operation for acoustically sensitive areas and uplink flexibility with two dual-personality ports that can be used as either RJ-45 Gigabit Ethernet or SFP ports
- Four built-in Gigabit Ethernet uplinks on 24- and 48- port models. Gigabit models have small form factor pluggable (SFP) for fiber connectivity and Fast Ethernet models have two SFP and two RJ-45 Gigabit uplinks
- IPv6
  - IPv6 host allows the switch to be deployed and managed at the edge of an IPv6 network
  - Dual stack (IPv4/IPv6) supports connectivity for both protocols; provides a transition mechanism from IPv4 to IPv6
  - MLD snooping forwards IPv6 multicast traffic to appropriate interface; prevents IPv6 multicast traffic from flooding the network
  - IPv6 ACL/QoS supports ACL and QoS for IPv6 network traffic on Gigabit and 48 port 10/100 models
  - Security RA Guard, DHCPv6 Protection, Dynamic IPv6 Lockdown (YA only)
- IEEE 802.3at Power over Ethernet (PoE+) provides up to 30 W per port that allows support of the latest PoE+ capable devices such as IP phones, wireless access points, and security cameras, as well as any IEEE 802.3af compliant end device; eliminates the cost of additional electrical cabling and circuits that would otherwise be necessary in IP phone and WLAN deployments
- Auto-MDIX adjusts automatically for straight-through or crossover cables on all ports
- Pre-standard PoE support detects and provides power to pre-standard PoE devices
- SFP slots provides fiber connectivity such as Gigabit-SX, LX, LH, and BX with four SFP slots on all 24- and 48-port Gigabit Ethernet models. Fast Ethernet 24- and 48-port models have two SFP slots and two RJ-45 Gigabit uplinks; 8-port models have two dual-personality ports supporting either SFP or RJ-45 Gigabit uplinks
- Dual-personality (RJ-45 or USB micro-B) serial console port gives easy access to switch CLI with front-of-switch location and the flexibility of using either an RJ-45 or USB micro-B serial console port

Layer 2 switching

- Support for 512 VLANs and 4,094 VLAN IDs
- Jumbo packet support improves the performance of large data transfers; supports frame size of up to 9,220 bytes; 8- and 24-port Fast Ethernet models automatically support up to 2,000-byte frames with no configuration needed
- 16K MAC address table provides access to many Layer 2 devices
- GARP VLAN Registration Protocol allows automatic learning and dynamic assignment of VLANs
- Rapid Per-VLAN Spanning Tree (RPVST+) allows each VLAN to build a separate spanning tree to improve link bandwidth usage; is compatible with PVST+

Security

- Access control lists (ACLs) accommodate IPv4/IPv6 port and VLAN-based ACLs (IPv6 ACL is supported only on Gigabit Ethernet and 48-port models.)
- Source-port filtering allows only specified ports to communicate with each other
- RADIUS/TACACS+ eases switch management security administration by using a password authentication server

- Secure Sockets Layer (SSL) encrypts all HTTP traffic, allowing secure access to the browser-based management GUI in the switch
- Port security allows access only to specified MAC addresses, which can be learned or specified by the administrator
- MAC address lockout prevents particular configured MAC addresses from connecting to the network
- Multiple user authentication methods
  - Uses an IEEE 802.1X supplicant on the client in conjunction with a RADIUS server to authenticate in accordance with industry standards
  - Web-based authentication provides a browser-based environment, similar to IEEE 802.1X, to authenticate clients that do not support the IEEE 802.1X supplicant
  - Supports MAC-based client authentication
- Secure shell (SSH) v2 encrypts all transmitted data for secure remote CLI access over IP networks
- STP BPDU port protection blocks Bridge Protocol Data Units (BPDUs) on ports that do not require BPDUs, preventing forged BPDU attacks
- STP root guard protects the root bridge from malicious attacks or configuration mistakes
- Secure management access delivers protected encryption of all access methods (CLI, GUI, or MIB) through SSHv2 and SNMPv3
- Custom banner displays security policy when users log in to the switch
- Secure FTP allows secure file transfer to and from the switch; protects against unwanted file downloads or unauthorized copying of a switch configuration file
- Protected ports CLI offers intuitive CLI to configure the source-port filter feature, by allowing specified ports to be isolated from all other ports on the switch; the protected port or ports can communicate only with the uplink or shared resources
- Authentication flexibility
  - Multiple IEEE 802.1X users per port provides authentication for up to 32 IEEE 802.1X users per port; prevents a user from "piggybacking" on another user's IEEE 802.1X authentication
  - Concurrent IEEE 802.1X, Web or MAC authentication schemes per port allows a switch port to accept IEEE 802.1X and either Web or MAC authentications
- Switch management logon security helps secure switch CLI logon by optionally requiring either RADIUS or TACACS+ authentication
- DHCP protection blocks DHCP packets from unauthorized DHCP servers, preventing denial-of-service attacks
- Dynamic ARP protection blocks ARP broadcasts from unauthorized hosts, preventing eavesdropping or theft of network data
- Dynamic IP lockdown works with DHCP protection to block traffic from unauthorized hosts, preventing IP source address spoofing
- MAC Pinning allows non-chatty legacy devices to stay authenticated by pinning client MAC addresses to the port until the clients logoff or get disconnected

Convergence

- IEEE 802.1AB Link Layer Discovery Protocol (LLDP) facilitates easy mapping using network management applications with LLDP automated device discovery protocol
- LLDP-MED (Media Endpoint Discovery) defines a standard extension of LLDP that stores values for parameters such as QoS and VLAN to automatically configure network devices such as IP phones
- PoE and PoE+ allocations support multiple methods (automatic, IEEE 802.3at dynamic, LLDP-MED fine grain, IEEE 802.3af device class or user-specified), to allocate and manage PoE/PoE+ power for more efficient energy savings
- Voice VLAN uses LLDP-MED to automatically configure a VLAN for IP phones
- IP multicast (IGMP) prevents flooding of IP multicast traffic
- LLDP-CDP compatibility receives and recognizes CDP packets from Cisco's IP phones for seamless interoperation
- Local MAC Authentication assigns attributes such as VLAN and QoS using locally configured profile that can be a list of MAC prefixes Unified Wired and Wireless

Resiliency and high availability

- Port trunking and link aggregation
  - Trunking supports up to eight links per trunk to increase bandwidth and create redundant connections; and supports L2, L3, and L4 trunk load-balancing algorithm (L4 trunk load balancing is supported only on Gigabit Ethernet and 48-port models.)
  - IEEE 802.3ad Link Aggregation Control Protocol (LACP) eases configuration of trunks through automatic configuration
- IEEE 802.1s Multiple Spanning Tree provides high link availability in multiple VLAN environments by allowing multiple spanning trees; provides legacy support for IEEE 802.1d and IEEE 802.1w
- SmartLink provides easy-to-configure link redundancy of active and standby links

Product architecture

- Energy-efficient design
  - IEEE 802.3az reduces power consumption during periods of low data activity on Gigabit Ethernet switches
  - Port low-power mode enables the port to automatically go into low-power mode to conserve energy when no link is detected
  - Fan-less and variable-speed fans decrease power consumption in fan-less (all 8-port, 2530-24, and 2530-48 PoE+ switches) as well as variable-speed fan switches
  - Port LEDs conserves energy by optionally turning off port link and activity LEDs
- Switch on a chip provides a highly integrated, high-performance switch design with a nonblocking architecture

Flexibility

- Power Savings with Energy Efficient Design
  - Rack mountable allows the switch to be mounted on a standard 19-inch rack, with the hardware included
  - Wall mountable allows the switch to be mounted on a wall, using the hardware included
  - Surface mountable allows the switch to be mounted above or below a surface (such as a desk or table), using the hardware included
- Quiet operation lowers noise, making it suitable for deployments in acoustically sensitive environments such as conference rooms and office spaces
- Compact size reduces space requirements (refer to the product specifications for the exact dimensions)

Warranty and support

- Limited Lifetime Warranty
  See **www.hpe.com/networking/warrantysummary** for warranty and support information included with your product purchase.
- To find software for your product, refer to **www.hpe.com/networking/support**; for details on the software releases available with your product purchase, refer to **www.hpe.com/networking/warrantysummary**
- Services
  Refer to the Hewlett Packard Enterprise website at **www.hpe.com/networking/services** for details on the service-level descriptions and product numbers. For details about services and response times in your area, please contact your local Hewlett Packard Enterprise sales office.

SPECIFICATIONS

	Aruba 2530-48G-PoE+ Switch (J9772A) (J9772ACM[1])	Aruba 2530-24G-PoE+ Switch (J9773A) (J9773ACM[1])	Aruba 2530-8G-PoE+ Switch (J9774A) (J9774ACM[1])
I/O ports and slots			
	48 RJ-45 autosensing 10/100/1000 PoE+ ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T, IEEE 802.3at PoE+); Media Type: Auto-MDIX; Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only	24 RJ-45 autosensing 10/100/1000 PoE+ ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T, IEEE 802.3at PoE+); Media Type: Auto-MDIX; Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only	8 RJ-45 autosensing 10/100/1000 PoE+ ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T, IEEE 802.3at PoE+); Media Type: Auto-MDIX; Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only
	4 fixed Gigabit Ethernet SFP ports	4 fixed Gigabit Ethernet SFP ports	2 dual-personality ports, each port can be used as either an RJ-45 10/100/1000 port (IEEE 802.3 Type 10Base-T; IEEE 802.3u Type 100Base-TX; IEEE 802.3ab1000Base-T Gigabit Ethernet) or as a SFP slot (for use with SFP transceivers)
	1 dual-personality (RJ-45 or USB micro-B) serial console port	1 dual-personality (RJ-45 or USB micro-B) serial console port	1 dual-personality (RJ-45 or USB micro-B) serial console port
Physical characteristics			
Dimensions	17.44 (w) x 13.00 (d) x 1.75 (h) in (44.3 x 32.26 x 4.45 cm) (1U height)	17.44 (w) x 13.00 (d) x 1.75 (h) in (44.3 x 33.02 x 4.45 cm) (1U height)	10.00(w) x 6.28(d) x 1.75(h) in (25.4 x 15.95 x 4.45 cm) (1U height)
Weight	10.4 lb (4.72 kg)	8.7 lb (3.95 kg)	2.2 lb (1 kg)
Memory and processor			
Processor	ARM9E @ 800 MHz, 128 MB flash, 256 MB DDR3 DIMM, packet buffer size: 3 MB dynamically allocated	ARM9E @ 800 MHz, 128 MB flash, 256 MB DDR3 DIMM; packet buffer size: 1.5 MB dynamically allocated	ARM9E @ 800 MHz, 128 MB flash, 256 MB DDR3 DIMM; packet buffer size: 1.5 MB dynamically allocated
Mounting and enclosure			
	Mounts in an EIA-standard 19-inch telco rack or equipment cabinet (rack-mounting kit available); Horizontal surface mounting; Wall mounting	Mounts in an EIA-standard 19-inch telco rack or equipment cabinet (rack-mounting kit available); Horizontal surface mounting; Wall mounting	Mounts in an EIA-standard 19-inch telco rack or equipment cabinet (rack-mounting kit available); Horizontal surface mounting; Wall mounting
Performance			
	IPv6 Ready Certified	IPv6 Ready Certified	IPv6 Ready Certified
100 Mb Latency	< 7.4 µs (LIFO 64-byte packets)	< 7.4 µs (LIFO 64-byte packets)	< 7.4 µs (LIFO 64-byte packets)
1000 Mb Latency	< 2.3 µs (LIFO 64-byte packets)	< 2.3 µs (LIFO 64-byte packets)	< 2.6 µs (LIFO 64-byte packets)
Throughput	up to 77.3 Mpps (64-byte packets)	up to 41.6 Mpps (64-byte packets)	up to 14.8 Mpps (64-byte packets)
Switching capacity	104 Gbps	56 Gbps	20 Gbps
MAC address table size	16000 entries	16000 entries	16000 entries

SPECIFICATIONS

	Aruba 2530-48G-PoE+ Switch (J9772A) (J9772ACM[1])	Aruba 2530-24G-PoE+ Switch (J9773A) (J9773ACM[1])	Aruba 2530-8G-PoE+ Switch (J9774A) (J9774ACM[1])
Environment			
Operating temperature	32°F to 113°F (0°C to 45°C)	32°F to 113°F (0°C to 45°C)	32°F to 113°F (0°C to 45°C)
Operating relative humidity	15% to 95% @ 104°F (40°C), noncondensing	15% to 95% @ 104°F (40°C), noncondensing	15% to 95% @ 104°F (40°C), noncondensing
Nonoperating/Storage temperature	-40°F to 158°F (-40°C to 70°C)	-40°F to 158°F (-40°C to 70°C)	-40°F to 158°F (-40°C to 70°C)
Nonoperating/Storage relative humidity	15% to 90% @ 149°F (65°C), noncondensing	15% to 90% @ 149°F (65°C), noncondensing	15% to 90% @ 149°F (65°C), noncondensing
Altitude	up to 10,000 ft (3 km)	up to 10,000 ft (3 km)	up to 10,000 ft (3 km)
Acoustic	Power: 43.6 dB, Pressure: 33.6 dB	Power: 43.9 dB, Pressure: 39.6 dB	Power: 0 dB, Pressure: 0 dB
Electrical characteristics			
Frequency	50/60 Hz	50/60 Hz	50/60 Hz
Maximum heat dissipation	236 BTU/hr (248.98 kJ/hr), (switch only: 236 BTU/hr; combined switch + max. PoE devices: 1624 BTU/hr)	135 BTU/hr (142.42 kJ/hr), (switch only: 135 BTU/hr; combined switch + max. PoE devices: 843 BTU/hr)	65 BTU/hr (68.58 kJ/hr), (switch only: 65 BTU/hr; combined switch + max. PoE devices: 293 BTU/hr)
AC voltage	100 - 127/200 - 240 VAC	100 - 127/200 - 240 VAC	100 - 127/200 - 240 VAC
Current	5.8/2.9 A	3.2/1.6 A	1.4 A
Maximum power rating	476 W	247 W	86 W
Idle power	40.1 W	25.2 W	13.4 W
PoE power	382 W	195 W	67 W
Notes	Idle power is the actual power consumption of the device with no ports connected.	Idle power is the actual power consumption of the device with no ports connected.	Idle power is the actual power consumption of the device with no ports connected.
	Maximum power rating and maximum heat dissipation are the worst-case theoretical maximum numbers provided for planning the infrastructure with fully loaded PoE (if equipped), 100% traffic, all ports plugged in, and all modules populated.	Maximum power rating and maximum heat dissipation are the worst-case theoretical maximum numbers provided for planning the infrastructure with fully loaded PoE (if equipped), 100% traffic, all ports plugged in, and all modules populated.	Maximum power rating and maximum heat dissipation are the worst-case theoretical maximum numbers provided for planning the infrastructure with fully loaded PoE (if equipped), 100% traffic, all ports plugged in, and all modules populated.
	PoE power is the total power budget available to all PoE ports.	PoE power is the total power budget available to all PoE ports.	PoE power is the total power budget available to all PoE ports.
Safety	UL 60950-1; CAN/CSA 22.2 No. 60950-1; EN 60825; IEC 60950-1; EN 60950-1	UL 60950-1; CAN/CSA 22.2 No. 60950-1; EN 60825; IEC 60950-1; EN 60950-1	UL 60950-1; CAN/CSA 22.2 No. 60950-1; EN 60825; IEC 60950-1; EN 60950-1
Emissions	FCC Class A; EN 55022/CISPR-22 Class A; VCCI Class A	FCC Class A; EN 55022/CISPR-22 Class A; VCCI Class A	FCC Class A; EN 55022/CISPR-22 Class A; VCCI Class A

SPECIFICATIONS

	Aruba 2530-48G-PoE+ Switch (J9772A) (J9772ACM[1])	Aruba 2530-24G-PoE+ Switch (J9773A) (J9773ACM[1])	Aruba 2530-8G-PoE+ Switch (J9774A) (J9774ACM[1])
Immunity			
Generic	EN 55024, CISPR 24	EN 55024, CISPR 24	EN 55024, CISPR 24
EN	EN 55024, CISPR 24	EN 55024, CISPR 24	EN 55024, CISPR 24
ESD	IEC 61000-4-2	IEC 61000-4-2	IEC 61000-4-2
Radiated	IEC 61000-4-3	IEC 61000-4-3	IEC 61000-4-3
EFT/Burst	IEC 61000-4-4	IEC 61000-4-4	IEC 61000-4-4
Surge	IEC 61000-4-5	IEC 61000-4-5	IEC 61000-4-5
Conducted	IEC 61000-4-6	IEC 61000-4-6	IEC 61000-4-6
Power frequency magnetic field	IEC 61000-4-8	IEC 61000-4-8	IEC 61000-4-8
Voltage dips and interruptions	IEC 61000-4-11	IEC 61000-4-11	IEC 61000-4-11
Harmonics	EN 61000-3-2, IEC 61000-3-2	EN 61000-3-2, IEC 61000-3-2	EN 61000-3-2, IEC 61000-3-2
Flicker	EN 61000-3-3, IEC 61000-3-3	EN 61000-3-3, IEC 61000-3-3	EN 61000-3-3, IEC 61000-3-3
Management			
	Aruba Central and Aruba AirWave Network Management; IMC—Intelligent Management Center; command-line interface; Web browser; configuration menu; out-of-band management (serial RS-232C or Micro USB); IEEE 802.3 Ethernet MIB; Repeater MIB; Ethernet Interface MIB AirWave Network Management	Aruba Central and Aruba AirWave Network Management; IMC—Intelligent Management Center; command-line interface; Web browser; configuration menu; out-of-band management (serial RS-232C or Micro USB); IEEE 802.3 Ethernet MIB; Repeater MIB; Ethernet Interface MIB AirWave Network Management	Aruba Central and Aruba AirWave Network Management, IMC—Intelligent Management Center; command-line interface; Web browser; configuration menu; out-of-band management (serial RS-232C or Micro USB); IEEE 802.3 Ethernet MIB; Repeater MIB; Ethernet Interface MIB AirWave Network Management
Notes			
	IEEE 802.3az applies to Gigabit models only; IEEE 802.3at and IEEE 802.3af apply to PoE+ models only. When using SFPs with this product, SFPs with revision "B" or later (product number ends with the letter "B" or later, e.g., J4858B, J4859C) are required.	IEEE 802.3az applies to Gigabit models only; IEEE 802.3at and IEEE 802.3af apply to PoE+ models only. When using SFPs with this product, SFPs with revision "B" or later (product number ends with the letter "B" or later, e.g., J4858B, J4859C) are required.	IEEE 802.3az applies to Gigabit models only; IEEE 802.3at and IEEE 802.3af apply to PoE+ models only. When using SFPs with this product, SFPs with revision "B" or later (product number ends with the letter "B" or later, e.g., J4858B, J4859C) are required.
Services			
	Refer to the Hewlett Packard Enterprise website at www.hpe.com/networking/services for details on the service-level descriptions and product numbers. For details about services and response times in your area, please contact your local Hewlett Packard Enterprise sales office.	Refer to the Hewlett Packard Enterprise website at www.hpe.com/networking/services for details on the service-level descriptions and product numbers. For details about services and response times in your area, please contact your local Hewlett Packard Enterprise sales office.	Refer to the Hewlett Packard Enterprise website at www.hpe.com/networking/services for details on the service-level descriptions and product numbers. For details about services and response times in your area, please contact your local Hewlett Packard Enterprise sales office.

STANDARDS AND PROTOCOLS
(APPLIES TO ALL PRODUCTS IN SERIES)

Denial of service protection
- Network DoS Filter

Device management
- RFC 1591 DNS (client)
- SSHv1/SSHv2 Secure Shell
- RFC 2576 (Coexistence between SNMP V1, V2, V3)
- RFC 2579 (SMIv2 Text Conventions)
- RFC 2580 (SMIv2 Conformance)
- RFC 3416 (SNMP Protocol Operations v2)
- RFC 3417 (SNMP Transport Mappings)

General protocols
- IEEE 802.1D MAC Bridges
- IEEE 802.1p Priority
- IEEE 802.1Q VLANs
- IEEE 802.1s Multiple Spanning Trees
- IEEE 802.1w Rapid Reconfiguration of Spanning Tree
- IEEE 802.3 Type 10BASE-T
- IEEE 802.3ab 1000BASE-T
- IEEE 802.3ad Link Aggregation Control Protocol (LACP)
- IEEE 802.3af Power over Ethernet
- IEEE 802.3at Power over Ethernet Plus
- IEEE 802.3az Energy Efficient Ethernet
- IEEE 802.3x Flow Control
- RFC 768 UDP
- RFC 783 TFTP Protocol (revision 2)
- RFC 792 ICMP
- RFC 793 TCP
- RFC 826 ARP
- RFC 854 TELNET
- RFC 868 Time Protocol
- RFC 951 BOOTP
- RFC 1350 TFTP Protocol (revision 2)
- RFC 1542 BOOTP Extensions
- RFC 1918 Address Allocation for Private Internet
- RFC 2030 Simple Network Time Protocol (SNTP) v4
- RFC 2131 DHCP
- RFC 3411 An Architecture for Describing Simple Network Management Protocol (SNMP) Management Frameworks
- RFC 3412 Message Processing and Dispatching for the Simple Network Management Protocol (SNMP)
- RFC 3413 Simple Network Management Protocol (SNMP) Applications
- RFC 3414 User-based Security Model (USM) for version 3 of the Simple Network Management Protocol (SNMPv3)
- RFC 3415 View-based Access Control Model (VACM) for the Simple Network Management Protocol (SNMP)
- RFC 3575 IANA Considerations for RADIUS
- RFC 5905 NTP Client

IP multicast
- RFC 2236 IGMPv2

IPv6
- RFC 1981 IPv6 Path MTU Discovery
- RFC 2460 IPv6 Specification
- RFC 2464 Transmission of IPv6 over Ethernet Networks
- RFC 2925 Remote Operations MIB (Ping only)
- RFC 3315 DHCPv6 (client only)
- RFC 3484 Default Address Selection for IPv6
- RFC 3513 IPv6 Addressing Architecture
- RFC 3596 DNS Extension for IPv6
- RFC 3810 Multicast Listener Discovery Version 2 (MLDv2) for IPv6
- RFC 4022 MIB for TCP
- RFC 4113 MIB for UDP
- RFC 4251 SSHv6 Architecture
- RFC 4252 SSHv6 Authentication
- RFC 4252 SSHv6 Transport Layer
- RFC 4254 SSHv6 Connection
- RFC 4291 IP Version 6 Addressing Architecture
- RFC 4293 MIB for IP
- RFC 4419 Key Exchange for SSH
- RFC 4443 ICMPv6
- RFC 4861 IPv6 Neighbor Discovery
- RFC 4862 IPv6 Stateless Address Auto-configuration
- RFC 5095 Deprecation of Type 0 Routing Headers in IPv6

MIBs
- RFC 1155 Structure and Identification of Management Information for TCP/IP Internets
- RFC 1212 Concise MIB Definitions
- RFC 1213 MIB II
- RFC 1493 Bridge MIB
- RFC 2021 RMONv2 MIB
- RFC 2578 Structure of Management Information Version 2 (SMIv2)
- RFC 2579 Textual Conventions for SMIv2
- RFC 2613 SMON MIB
- RFC 2618 RADIUS Client MIB
- RFC 2620 RADIUS Accounting Client MIB
- RFC 2665 Ethernet-Like-MIB 2
- RFC 2668 802.3 MAU MIB
- RFC 2674 802.1p and IEEE 802.1Q Bridge MIB
- RFC 2737 Entity MIB (Version 2)
- RFC 2863 The Interfaces Group MIB
- RFC 4836 Managed Objects for 802.3 Medium Attachment Units (MAU)

Network management

- IEEE 802.1AB Link Layer Discovery Protocol (LLDP)
- RFC 1098 A Simple Network Management Protocol (SNMP)
- RFC 1155 Structure of Management Information
- RFC 2819 Four groups of RMON: 1 (statistics), 2 (history) 3 (alarm) and 9 (events)
- RFC 3411 SNMP Management Frameworks
- RFC 3412 Message Processing and Dispatching for the Simple Network Management Protocol (SNMP)
- RFC 3413 Simple Network Management Protocol (SNMP) Applications
- RFC 3414 User-based Security Model (USM) for version 3 of the Simple Network Management Protocol (SNMPv3)
- RFC 3415 View-based Access Control Model (VACM) for the Simple Network Management Protocol (SNMP)
- RFC 3418 Management Information Base (MIB) for the Simple Network Management Protocol (SNMP)
- RFC 5424 Syslog Protocol
- ANSI/TIA-1057 LLDP Media Endpoint Discovery (LLDP-MED)
- SNMPv1/v2c/v3

QoS/CoS

- RFC 2474 DiffServ precedence, with 4 queues per port
- RFC 2475 DiffServ Architecture
- RFC 2597 DiffServ Assured Forwarding (AF)
- RFC 2598 DiffServ Expedited Forwarding (EF)

Security

- IEEE 802.1X Port Based Network Access Control
- RFC 1492 TACACS+
- RFC 2138 RADIUS Authentication
- RFC 2866 RADIUS Accounting
- RFC 7030 Enrollment over Secure Transport
- Secure Sockets Layer (SSL)

ARUBA 2530 SWITCHES AND ACCESSORIES

Switch Models

- Aruba 2530-48G-PoE+ Switch (J9772A)
- Aruba CM 2530-48G-PoE+ Switch (J9772ACM)[1]
- Aruba 2530-24G-PoE+ Switch (J9773A)
- Aruba CM 2530-24G-PoE+ Switch (J9773ACM)[1]
- Aruba 2530-8G-PoE+ Switch (J9774A)
- Aruba CM 2530-8G-PoE+ Switch (J9774ACM)
- Aruba 2530-48G Switch (J9775A)
- Aruba 2530-24G Switch (J9776A)
- Aruba 2530-8G Switch (J9777A)
- Aruba 2530-48-PoE+ Switch (J9778A)
- Aruba 2530-24-PoE+ Switch (J9779A)
- Aruba 2530-8-PoE+ Switch (J9780A)
- Aruba 2530-48 Switch (J9781A)
- Aruba 2530-24 Switch (J9782A)
- Aruba 2530-8 Switch (J9783A)
- Aruba 2530-8-PoE+ Internal PS Switch (JL070A)

Transceivers

- Aruba 100M SFP LC FX 2km MMF XCVR (J9054D)
- Aruba CM 100M SFP LC FX 2km MMF XCVR (J9054DCM)
- Aruba 1G SFP RJ45 T 100m Cat5e XCVR (J8177D)
- Aruba CM 1G SFP RJ45 T 100m Cat5e XCVR (J8177DCM)[1]
- Aruba 1G SFP LC SX 500m MMF XCVR (J4858D)
- Aruba CM 1G SFP LC SX 500m MMF XCVR (J4858DCM)[1]
- Aruba 1G SFP LC LX 10km SMF XCVR (J4859D)
- Aruba CM 1G SFP LC LX 10km SMF XCVR (J4859DCM)[1]
- Aruba 1G SFP LC LH 70km SMF XCVR (J4860D)
- Aruba CM 1G SFP LC LH 70km SMF XCVR (J4860DCM)[1]

Cables

- Aruba X2C2 RJ45 to DB9 Console Cable (JL448A)

Mounting Kit

- HPE X410 1U Universal 4 post Rack Mounting Kit (J9583A)

Aruba 2530-8-PoE+ Internal PS Switch (JL070A)

- HPE X510 1U Cable Guard (J9700A)

Aruba 2530-8G-PoE+ Switch (J9774A)

- Aruba 2530 8-port Switch Power Adapter Shelf (J9820A)
- Aruba CM 2530 8-port Switch Power Adapter Shelf (J9820ACM)[1]
- Aruba X510 1U Cable Guard (J9700A)
- Aruba CM X510 1U Cable Guard (J9700ACM)[1]

Aruba 2530-8-PoE+ Switch (J9780A)

- Aruba 2530 8-port Switch Power Adapter Shelf (J9820A)
- Aruba X510 1U Cable Guard (J9700A)

Aruba 2530-8G Switch (J9777A)

- Aruba 2530 8-port Switch Power Adapter Shelf (J9820A)
- Aruba X510 1U Cable Guard (J9700A)

Aruba 2530-8 Switch (J9783A)

- Aruba 2530 8-port Switch Power Adapter Shelf (J9820A)
- HPE X510 1U Cable Guard (J9700A)

[1] All hardware SKUs can be managed by Aruba Central. Central Managed (CM) SKUs are used for simplified ordering within U.S. and Canada only. Append 'CM' to the indicated SKU #: (e.g., J9772ACM to order the J9772A). Requires an active Central license and end-user information consistent with the Central license purchase. Applicable accessories with a valid 'CM' suffix should also be placed on the same order.

aruba
a Hewlett Packard Enterprise company

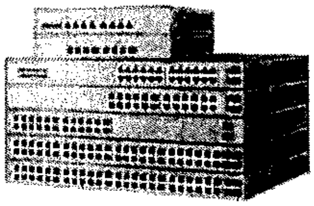
DS_2530_SK_101819 4AA4-4245ENW

2.2. PoE switch 24 port

Hewlett Packard Enterprise

Switch POE 24Port - JL385A

HPE OfficeConnect 1920S Switch Series



Key features

- Customized operation using intuitive Web interface
- Layer 3 static routing with 32 routes for network segmentation and expansion
- Access control lists for granular security control
- Spanning Tree Protocol: STP, RSTP, and MSTP
- 8-, 24- and 48-port non-PoE+ models are fanless for quiet operation
- HPE Limited Lifetime Warranty

Product overview

The HPE OfficeConnect 1920S Switch Series consists of advanced smart-managed fixed-configuration Gigabit switches designed for small businesses in an easy-to-administer solution.

The series consists of seven switches including 8- 24- and 48-port Gigabit Ethernet switches and 8-, 24- and 48-port PoE+ models of which half the ports are POE+ capable. An additional 24-port PoE+ model is available that provides PoE+ on all 24-ports. All ports provide non-blocking Gigabit performance. Some models include SFP ports for fiber connectivity and the 8-, 24- and 48-port non PoE+ models are fanless, making them ideal for office deployments. All HPE OfficeConnect 1920S Switches support flexible installation options, including mounting on wall, under table, or on desktop. The 8-port Gigabit Ethernet model can be powered by an upstream Power over Ethernet switch for environments where no line power is available.

The series is part of the OfficeConnect portfolio of Hewlett Packard Enterprise small business networking products. These switches provide a great value, and includes features to satisfy even the most advanced small business networks. Customizable features include basic Layer 2 features like VLANs and link aggregation, as well as advanced features such as Layer 3 static routing, IPv4 and IPv6 Host mode, ACLs, and Spanning Tree Protocols. HPE OfficeConnect 1920S Switch Series includes a Limited Lifetime Warranty.

Features and benefits

Management

Simple Web management
Allows for easy management of the switch—even by nontechnical users—through an intuitive Web GUI; supports HTTP and HTTP Secure (HTTPS).

SNMPv1, v2c, and v3
Facilitate management of the switch, as the device can be discovered and monitored from an SNMP management station.

Complete session logging
Provides detailed information for problem identification and resolution.

Port mirroring
Enables traffic on a port or VLAN to be simultaneously sent to a network analyzer for monitoring.

Dual flash images
Provide independent primary and secondary operating system files for backup while upgrading.

Network Time Protocol (NTP)
Synchronizes timekeeping among distributed time servers; keeps timekeeping consistent among all clock-dependent devices within the network so that the devices can provide diverse applications based on the consistent time.

Manual network time configuration
Manually set the date and time on the switch in the absence of an NTP server.

Default DHCP client mode
Allows the switch to be directly connected to a network, enabling plug-and-play operation; in absence of a DHCP server on the network, the switch falls back to the static address 192.168.1.1

FTP and TFTP
Provides different mechanisms for configuration updates; FTP allows bidirectional transfers over a TCP/IP network; trivial FTP (TFTP) is a simpler method using user Datagram Protocol (UDP).

Remote monitoring (RMON)
Remote monitoring (RMON) provides advanced monitoring and reporting capabilities for statistics, history, alarms and events. RMON data is retrieved from the switch through a network management platform over SNMP.

Quality of service (QoS)
Traffic prioritization
Provides time-sensitive packets (like VoIP and video) with priority over other traffic based on DSCP or IEEE 802.1p classification.

IEEE 802.1p/Q VLAN tagging
Delivers data to devices based on the priority and type of traffic; supports IEEE 802.1Q.

Advanced classifier based QoS
Classifies traffic using multiple match criteria based on Layer 2, 3, and 4 information.

Packet storm protection
Protects against unknown unicast, broadcast and multicast storms with user-defined thresholds.

Rate limiting
Sets per-port ingress enforced maximum or percent minimum bandwidth per queue.

Class of Service (CoS)
Sets the IEEE 802.1p priority tag based on IP address, IP Type of Service (ToS), Layer 3 protocol, TCP/UDP port number or source port.

Powerful QoS feature
Supports the following congestion actions: strict priority queuing (SP) or weighted round robin (WRR) queuing. SP and WRR queuing can be configured on individual switch ports.

Connectivity
IPv6 host
Enables switches to be managed and deployed at the IPv6 network's edge.

IEEE 802.3X Flow Control
Provides a flow throttling mechanism propagated through the network to prevent packet loss at a congested node.

IEEE 802.3at Power over Ethernet (PoE+)
Provides up to 30 W per port, which allows support of the latest PoE+ capable devices such as Video IP phones, wireless access points, and advanced pan/tilt/zoom security cameras, as well as any 15.4 W IEEE 802.3af-compliant end device; mitigates the cost of additional electrical cabling and circuits that would otherwise be necessary in IP phone and WLAN deployments.

PoE+ port availability
Ports 1–4 are PoE/PoE+ capable on the HPE OfficeConnect 1920S 8G PPoE+ 65W switch; ports 1–12 are PoE/PoE+ capable on the HPE OfficeConnect 1920S 24G 2SFP PPoE+ 185W switch; all ports provide PoE/PoE+ on the HPE OfficeConnect 1920S 24G 2SFP PoE+ 370W switch; ports 1–24 are PoE/PoE+ capable on the HPE OfficeConnect 1920S 48G 4SFP PPoE+ 370W switch.

Auto-PoE power configuration
The switch automatically assigns the required power to a port for a PD device based on Link Layer Discovery Protocol (LLDP). Optionally, the switch permits manual, per port, PoE power configuration.

PoE shut down mode
A PoE scheduler provides the ability to define the hours of PoE power being supplied to a group of switch ports based on a 24-hour day. The scheduler enables the flexibility to select individual days of a week as well as reoccurrence on a weekly basis with a start and end date.

PoE power allocation
Support multiple methods (automatic, IEEE 802.3af class, LLDP-MED, or user-specified) to allocate PoE power for more efficient energy savings.

SPF ports for fiber connectivity
Provides fiber connections for uplinks and other connections across longer distances than copper cabling can support; SFP ports are in addition to available copper Ethernet ports, providing a higher total number of available ports. Two SFP ports available on 24 and four SFP ports on 48 port models.

Loop protection
If the switch detects a loop, it disables the source port from forwarding data packets originating from the switch to avoid broadcast storms.

Auto MDI/MDI-X
Adjusts automatically for straight-through or crossover cables on all 10/100/1000 ports.

Energy Efficient Ethernet (EEE)
Compliant with IEEE 802.3az standard requirements to save energy during periods of low data activity.

Auto-port shut down
The switch saves power by automatically shutting down power to inactive ports. Power is restored on a port upon link detection.

Energy savings status
The switch provides an estimated cumulative energy savings due to green Ethernet features being enabled.

Energy-efficient cooling
Includes variable speed fans operating only at the speed necessary to maintain operating temperature to reduce excess noise and power consumption by the switch.

Security

Access Control Lists (ACLs)
Enables network traffic filtering by creating an ACL, add rules and match criteria to an ACL, and apply the ACL to permit or deny on one or more interfaces or a VLAN. Up to 50 inbound entries may be configured based on IPv4 source and destination IP and MAC address, Layer 4 ports and protocol type of the IPv4 packet.

RADIUS
The switch support RADIUS authentication and configuration of up to 8 RADIUS servers.

RADIUS Accounting
A robust set of attributes and statistics are available for collecting information from the switch.

IEEE 802.1X access control
Authentication of network users on a per port basis prior to permitting network access. Port VLAN includes RADIUS VLAN assignment, dynamic VLAN creation, guest VLAN or into an unauthenticated VLAN.

Switch 802.1X supplicant
Enables the switch to authenticate itself to a RADIUS server.

Port isolation
Ports in a port isolation group are restricted from forwarding Layer 2 traffic between ports in that group; provides data privacy and security.

Automatic denial-of-service protection
Monitors for malicious attacks and protects the network by blocking the attacks.

Management password
Provides security so that only authorized access to the Web browser interface is allowed.

Secure Sockets Layer (SSL)
Encrypts all HTTP traffic, secure access to the browser-based management of the switch.

Performance
Half-and full-duplex auto-negotiating capability on every port doubles the throughput of every port.

Selectable queue configurations
Allows for increased performance by selecting the number of queues and associated memory buffering that best meet the requirements of the network applications.

IGMP snooping
Improves network performance through multicast filtering, instead of flooding traffic on all ports.

SFP fiber uplinks
Provides greater distance connectivity using Gigabit fiber uplinks.

Layer 2 switching

Spanning Tree Protocol (STP)
Supports standard IEEE 802.1D STP, IEEE 802.1w Rapid Spanning Tree Protocol (RSTP) for faster convergence, and IEEE 802.1s Multiple Spanning Tree Protocol (MSTP).

BPDU filtering
Drops BPDU packets when STP is enabled globally but disabled on a specific port

Jumbo frame support
Supports up to 9216 bytes frame size to improve the performance of large data transfers.

VLAN support and tagging
Support for IEEE 802.1Q; 256 VLANs with a VLAN ID range of 2-4093.

Layer 3 services

Address Resolution Protocol (ARP)
Displays the MAC address of another IP host in the same subnet; supports static ARPs; proxy ARP allows normal ARP operation between subnets or when subnets are separated by a Layer 2 network.

DHCP Relay
Simplifies management of DHCP addresses in networks with multiple subnets.

Layer 3 routing

Static IPv4 routing
Provides basic routing supporting up to 32 static routes to allow manual routing configuration.

Link aggregation
Groups together multiple ports up to a maximum of eight ports per trunk either automatically using Link Aggregation Control Protocol (LACP), or manually, to form an ultra-high-bandwidth connection to the network backbone; help prevent traffic bottlenecks. The 8 port models support 4 trunks, 16 and 24 port models support 8 trunks, 48 port models support 16 trunks.

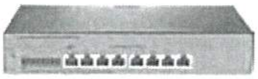
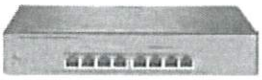
Convergence

LLDP-MED (Media Endpoint Discovery)
Defines a standard extension of LLDP that stores values for parameters such as QoS and VLAN to configure network devices such as IP phones automatically.

Auto-voice VLAN
Recognizes IP phones and automatically assigns voice traffic to dedicated VLAN for IP phones.

Warranty and support
See **hpe.com/officeconnect/support** for warranty and support information included with your product purchase.

HPE 1920S Switch Series

			
Specifications	**HPE OfficeConnect 1920S 8G Switch (JL380A)**	**HPE OfficeConnect 1920S 8G PPoE+ 65W Switch (JL383A)**	**HPE OfficeConnect 1920S 24G 2SFP Switch (JL381A)**
I/O ports and slots	8 RJ-45 autosensing 10/100/1000 ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T); Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only	4 RJ-45 autosensing 10/100/1000 PoE+ ports; Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 4 RJ-45 autosensing 10/100/1000 ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T); Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only	24 RJ-45 autosensing 10/100/1000 ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T); Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 2 SFP 100/1000 Mbps ports (IEEE 802.3z Type 1000BASE-X, IEEE 802.3u Type 100BASE-FX)
Physical characteristics			
Dimensions	10(w) x 6.28(d) x 1.73(h) in (25.4 x 15.95 x 4.39 cm) (1U height)	10(w) x 6.28(d) x 1.73(h) in (25.4 x 15.95 x 4.39 cm) (1U height)	17.42(w) x 9.69(d) x 1.73(h) in (44.25 x 24.61 x 4.39 cm) (1U height)
Weight	1.81 lb (0.82 kg)	2.01 lb (0.91 kg)	6 lb (2.72 kg)
Memory and processor	ARM Cortex-A9 @ 400 MHz, 256 MB SDRAM, 64 MB flash; packet buffer: 1.5 MB	ARM Cortex-A9 @ 400 MHz, 256 MB SDRAM, 64 MB flash; packet buffer: 1.5 MB	ARM Cortex-A9 @ 400 MHz, 256 MB SDRAM, 64 MB flash; packet buffer: 1.5 MB
Mounting and enclosure			Mounts in an EIA standard 19-inch telco rack or equipment cabinet (hardware included)
Performance			
100 Mb Latency	< 7µs	< 7µs	< 7µs
1000 Mb Latency	< 2.4µs	< 2.3µs	< 2µs
Throughput	Up to 11.9 Mpps (64-byte packets)	Up to 11.9 Mpps (64-byte packets)	Up to 38.6 Mpps (64-byte packets)
Routing/Switching capacity	16 Gbps	16 Gbps	52 Gbps
Routing table size	32 entries	32 entries	32 entries
MAC address table size	8000 entries	8000 entries	8000 entries
Reliability			
MTBF (years)	144.9	112.4	80.0
Environment			
Operating temperature	32°F to 104°F (0°C to 40°C)	32°F to 104°F (0°C to 40°C)	32°F to 104°F (0°C to 40°C)
Operating relative humidity	15% to 95%, noncondensing @ 104°F (40°C)	15% to 95%, noncondensing @ 104°F (40°C)	15% to 95%, noncondensing @ 104°F (40°C)
Nonoperating/Storage temperature	-40°F to 158°F (-40°C to 70°C)	-40°F to 158°F (-40°C to 70°C)	-40°F to 158°F (-40°C to 70°C)
Nonoperating/Storage relative humidity	15% to 95%, noncondensing @ 140°F (60°C)	15% to 95%, noncondensing @ 140°F (60°C)	15% to 95%, noncondensing @ 140°F (60°C)
Altitude	up to 10,000 ft (3 km)	up to 10,000 ft (3 km)	up to 10,000 ft (3 km)
Acoustic	Pressure: 0 dB No Fan	Pressure: 0 dB No Fan	Pressure: 0 dB No Fan
Electrical characteristics			
Frequency	50/60 Hz	50/60 Hz	50/60 Hz
AC voltage	100 - 240 VAC	100 - 240 VAC	100 - 127/200 - 240 VAC
Current	2 A	.9 A	.5/.3 A
Maximum power rating	9.5 W	72.9 W	15.7 W
Idle power	8.2 W	9.7 W	11.6 W
PoE power		65 W PoE+	
	Notes: Maximum power rating is the worst-case theoretical maximum value provided for planning the infrastructure with 100% traffic, all ports plugged in.	Maximum power rating is the worst-case theoretical maximum value for planning the infrastructure with fully loaded PoE, 100% traffic and all ports plugged in.	Maximum power rating is the worst-case theoretical maximum value provided for planning the infrastructure with 100% traffic, all ports plugged in.

HPE 1920S Switch Series (continued)

Specifications	HPE OfficeConnect 1920S 8G Switch (JL380A)	HPE OfficeConnect 1920S 8G PPoE+ 65W Switch (JL383A)	HPE OfficeConnect 1920S 24G 2SFP Switch (JL381A)
Safety	UL 60950-1; IEC 60950-1; EN 60950-1; CAN/CSA-C22.2 No. 60950-1; EN 60825-1	UL 60950-1, IEC 60950-1; EN 60950-1; CAN/CSA-C22.2 No. 60950-1; EN 60825-1	UL 60950-1; IEC 60950-1; EN 60950-1; CAN/CSA-C22.2 No. 60950-1; EN 60825-1
Emissions	VCCI Class A; CNS 13438; ICES-003 Issue 5 Class A; FCC CFR 47 Part 15, Class A; EN 55032: 2015/CISPR-32	VCCI Class A; CNS 13438; ICES-003 Issue 5 Class A; FCC CFR 47 Part 15, Class A; EN 55032: 2015/CISPR-32	VCCI Class A; CNS 13438; ICES-003 Issue 5 Class A; FCC CFR 47 Part 15, Class A; EN 55032: 2015/CISPR-32
Immunity			
Generic	EN 55024, CISPR 24	EN 55024, CISPR 24	EN 55024, CISPR 24
EN	EN 55024, CISPR 24	EN 55024, CISPR 24	EN 55024, CISPR 24
ESD	IEC 61000-4-2	IEC 61000-4-2	IEC 61000-4-2
Radiated	IEC 61000-4-3	IEC 61000-4-3	IEC 61000-4-3
EFT/Burst	IEC 61000-4-4	IEC 61000-4-4	IEC 61000-4-4
Surge	IEC 61000-4-5	IEC 61000-4-5	IEC 61000-4-5
Conducted	IEC 61000-4-6	IEC 61000-4-6	IEC 61000-4-6
Power frequency magnetic field	IEC 61000-4-8	IEC 61000-4-8	IEC 61000-4-8
Voltage dips and interruptions	IEC 61000-4-11	IEC 61000-4-11	IEC 61000-4-11
Harmonics	EN 61000-3-2, IEC 61000-3-2	EN 61000-3-2, IEC 61000-3-2	EN 61000-3-2, IEC 61000-3-2
Flicker	EN 61000-3-3, IEC 61000-3-3	EN 61000-3-3, IEC 61000-3-3	EN 61000-3-3, IEC 61000-3-3
Management	Web browser; SNMP Manager	Web browser; SNMP Manager	Web browser; SNMP Manager
Notes			Use only supported genuine HPE mini-GBICs with your switch.
Services	Refer to the Hewlett Packard Enterprise website at **hpe.com/networking/services** for details on the service-level descriptions and product numbers. For details about services, and response times in your area, please contact your local Hewlett Packard Enterprise sales office.	Refer to the Hewlett Packard Enterprise website at **hpe.com/networking/services** for details on the service-level descriptions and product numbers. For details about services, and response times in your area, please contact your local Hewlett Packard Enterprise sales office.	Refer to the Hewlett Packard Enterprise website at **hpe.com/networking/services** for details on the service-level descriptions and product numbers. For details about services, and response times in your area, please contact your local Hewlett Packard Enterprise sales office.

HPE OfficeConnect 1920S Switch Series



Specifications	HPE OfficeConnect 1920S 24G 2SFP PPoE+ 185W Switch (JL384A)	HPE OfficeConnect 1920S 24G 2SFP PoE+ 370W Switch (JL385A)	HPE OfficeConnect 1920S 48G 4SFP Switch (JL382A)
I/O ports and slots	12 RJ-45 autosensing 10/100/1000 PoE+ ports; Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 12 RJ-45 autosensing 10/100/1000 ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T); Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 2 SFP 100/1000 Mbps ports (IEEE 802.3z Type 1000BASE-X, IEEE 802.3u Type 100BASE-FX)	24 RJ-45 autosensing 10/100/1000 PoE+ ports; Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 2 SFP 100/1000 Mbps ports (IEEE 802.3z Type 1000BASE-X, IEEE 802.3u Type 100BASE-FX)	48 RJ-45 autosensing 10/100/1000 ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T); Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 4 SFP 100/1000 Mbps ports (IEEE 802.3z Type 1000BASE-X, IEEE 802.3u Type 100BASE-FX)
Physical characteristics			
Dimensions	17.42(w) x 9.69(d) x 1.73(h) in (44.25 x 24.61 x 4.39 cm) (1U height)	17.42(w) x 12.7(d) x 1.73(h) in (44.25 x 32.26 x 4.39 cm) (1U height)	17.42(w) x 9.69(d) x 1.73(h) in (44.25 x 24.61 x 4.39 cm) (1U height)
Weight	7.3 lb (3.31 kg)	9.7 lb (4.4 kg)	7.3 lb (3.31 kg)
Memory and processor	ARM Cortex-A9 @ 400 MHz, 256 MB SDRAM, 64 MB flash; packet buffer: 1.5 MB	ARM Cortex-A9 @ 400 MHz, 256 MB SDRAM, 64 MB flash; packet buffer: 1.5 MB	ARM Cortex-A9 @ 400 MHz, 256 MB SDRAM, 64 MB flash; packet buffer: 1.5 MB
Mounting and enclosure	Mounts in an EIA standard 19-inch telco rack or equipment cabinet (hardware included)	Mounts in an EIA standard 19-inch telco rack or equipment cabinet (hardware included)	Mounts in an EIA standard 19-inch telco rack or equipment cabinet (hardware included)
Performance			
100 Mb Latency	< 7µs	< 7µs	< 7µs
1000 Mb Latency	< 2µs	< 2µs	< 2µs
Throughput	Up to 38.6 Mpps (64-byte packets)	Up to 77.3 Mpps (64-byte packets)	Up to 77.3 Mpps (64-byte packets)
Routing/Switching capacity	52 Gbps	52 Gbps	104 Gbps
Routing table size	32 entries	32 entries	32 entries
MAC address table size	8000 entries	16000 entries	16000 entries
Reliability			
MTBF (years)	64.5	57.1	61.7
Environment			
Operating temperature	32°F to 104°F (0°C to 40°C)	32°F to 104°F (0°C to 40°C)	32°F to 104°F (0°C to 40°C)
Operating relative humidity	15% to 95%, noncondensing @ 104°F (40°C)	15% to 95%, noncondensing @ 104°F (40°C)	15% to 95%, noncondensing @ 104°F (40°C)
Nonoperating/Storage temperature	-40°F to 158°F (-40°C to 70°C)	-40°F to 158°F (-40°C to 70°C)	-40°F to 158°F (-40°C to 70°C)
Nonoperating/Storage relative humidity	15% to 95%, noncondensing @ 140°F (60°C)	15% to 95%, noncondensing @ 140°F (60°C)	15% to 95%, noncondensing @ 140°F (60°C)
Altitude	up to 10,000 ft (3 km)	up to 10,000 ft (3 km)	up to 10,000 ft (3 km)
Acoustic	Power: 36 dB	Power: 45 dB	Pressure: 0 dB No Fan
Electrical characteristics			
Frequency	50/60 Hz	50/60 Hz	50/60 Hz
AC voltage	100 - 127/200 - 240 VAC	100 - 127/200 - 240 VAC	100 - 127/200 - 240 VAC
Current	2.6/1.3 A	3.5/1.9 A	.8/.5 A
Maximum power rating	207.9 W	435 W	32.2 W
Idle power	19 W	34.2 W	23.3 W
PoE power	185 W PoE+	370 W PoE+	
	Notes: Maximum power rating is the worst-case theoretical maximum value provided for planning the infrastructure with 100% traffic, all ports plugged in.	Maximum power rating is the worst-case theoretical maximum value for planning the infrastructure with fully loaded PoE, 100% traffic and all ports plugged in.	Maximum power rating is the worst-case theoretical maximum value for planning the infrastructure with 100% traffic and all ports plugged in.

HPE OfficeConnect 1920S Switch Series (continued)

Specifications	HPE OfficeConnect 1920S 8G Switch (JL380A)	HPE OfficeConnect 1920S 8G PPoE+ 65W Switch (JL383A)	HPE OfficeConnect 1920S 24G 2SFP Switch (JL381A)
Safety	UL 60950-1; IEC 60950-1; EN 60950-1; CAN/CSA-C22.2 No. 60950-1; EN 60825-1	UL 60950-1; IEC 60950-1; EN 60950-1; CAN/CSA-C22.2 No. 60950-1; EN 60825-1	UL 60950-1; IEC 60950-1; EN 60950-1; CAN/CSA-C22.2 No. 60950-1; EN 60825-1
Emissions	VCCI Class A; CNS 13438; ICES-003 Issue 5 Class A; FCC CFR 47 Part 15, Class A; EN 55032: 2015/CISPR-32	VCCI Class A; CNS 13438; ICES-003 Issue 5 Class A; FCC CFR 47 Part 15, Class A; EN 55032: 2015/CISPR-32	VCCI Class A; CNS 13438; ICES-003 Issue 5 Class A; FCC CFR 47 Part 15, Class A; EN 55032: 2015/CISPR-32
Immunity			
Generic	EN 55024, CISPR 24	EN 55024, CISPR 24	EN 55024, CISPR 24
EN	EN 55024, CISPR 24	EN 55024, CISPR 24	EN 55024, CISPR 24
ESD	IEC 61000-4-2	IEC 61000-4-2	IEC 61000-4-2
Radiated	IEC 61000-4-3	IEC 61000-4-3	IEC 61000-4-3
EFT/Burst	IEC 61000-4-4	IEC 61000-4-4	IEC 61000-4-4
Surge	IEC 61000-4-5	IEC 61000-4-5	IEC 61000-4-5
Conducted	IEC 61000-4-6	IEC 61000-4-6	IEC 61000-4-6
Power frequency magnetic field	IEC 61000-4-8	IEC 61000-4-8	IEC 61000-4-8
Voltage dips and interruptions	IEC 61000-4-11	IEC 61000-4-11	IEC 61000-4-11
Harmonics	EN 61000-3-2, IEC 61000-3-2	EN 61000-3-2, IEC 61000-3-2	EN 61000-3-2, IEC 61000-3-2
Flicker	EN 61000-3-3, IEC 61000-3-3	EN 61000-3-3, IEC 61000-3-3	EN 61000-3-3, IEC 61000-3-3
Management	Web browser; SNMP Manager	Web browser; SNMP Manager	Web browser; SNMP Manager
Notes	Use only supported genuine HPE mini-GBICs with your switch.	Use only supported genuine HPE mini-GBICs with your switch.	Use only supported genuine HPE mini-GBICs with your switch.
Services	Refer to the Hewlett Packard Enterprise website at **hpe.com/networking/services**, for details on the service-level descriptions and product numbers. For details about services, and response times in your area, please contact your local Hewlett Packard Enterprise sales office.	Refer to the Hewlett Packard Enterprise website at **hpe.com/networking/services**, for details on the service-level descriptions and product numbers. For details about services, and response times in your area, please contact your local Hewlett Packard Enterprise sales office.	Refer to the Hewlett Packard Enterprise website at **hpe.com/networking/services**, for details on the service-level descriptions and product numbers. For details about services, and response times in your area, please contact your local Hewlett Packard Enterprise sales office.

HPE OfficeConnect 1920S Switch Series (continued)

Specifications	HPE OfficeConnect 1920S 48G 4SFP PPoE+ 370W Switch (JL386A)
I/O ports and slots	24 RJ-45 autosensing 10/100/1000 PoE+ ports; Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 24 RJ-45 autosensing 10/100/1000 ports (IEEE 802.3 Type 10BASE-T, IEEE 802.3u Type 100BASE-TX, IEEE 802.3ab Type 1000BASE-T): Duplex: 10BASE-T/100BASE-TX: half or full; 1000BASE-T: full only 4 SFP 100/1000 Mbps ports (IEEE 802.3z Type 1000BASE-X, IEEE 802.3u Type 100BASE-FX)
Physical characteristics	
Dimensions	17.42(w) x 12.7(d) x 1.73(h) in (44.25 x 32.26 x 4.39 cm) (1U height)
Weight	9.7 lb (4.4 kg)
Memory and processor	ARM Cortex-A9 @ 400 MHz, 256 MB SDRAM, 64 MB flash; packet buffer: 1.5 MB
Mounting and enclosure	Mounts in an EIA standard 19-inch telco rack or equipment cabinet (hardware included)
Performance	
100 Mb Latency	< 7µs
1000 Mb Latency	< 2µs
Throughput	Up to 77.3 Mpps (64-byte packets)
Routing/Switching capacity	104 Gbps
Routing table size	32 entries
MAC address table size	16000 entries
Reliability	
MTBF (years)	45
Environment	
Operating temperature	32°F to 104°F (0°C to 40°C)
Operating relative humidity	15% to 95%, noncondensing @ 104°F (40°C)
Nonoperating/Storage temperature	-40°F to 158°F (-40°C to 70°C)
Nonoperating/Storage relative humidity	15% to 95%, noncondensing @ 140°F (60°C)
Altitude	up to 10,000 ft (3 km)
Acoustic	Power: 45 dB

HPE OfficeConnect 1920S Switch Series (continued)

Specifications	HPE OfficeConnect 1920S 48G 4SFP PPoE+ 370W Switch (JL386A)
Electrical characteristics	
Frequency	50/60 Hz
AC voltage	100 - 127/200 - 240 VAC
Current	5.1/2.6 A
Maximum power rating	481 W
Idle power	54.8 W
PoE power	370 W PoE+
	Notes: Maximum power rating is the worst-case theoretical maximum value provided for planning the infrastructure with 100% traffic, all ports plugged in.
Safety	UL 60950-1; IEC 60950-1; EN 60950-1; CAN/CSA-C22.2 No. 60950-1; EN 60825-1
Emissions	VCCI Class A; CNS 13438; ICES-003 Issue 5; Class A; FCC CFR 47 Part 15, Class A; EN 55032: 2015/CISPR-32
Immunity	
Generic	EN 55024, CISPR 24
EN	EN 55024, CISPR 24
ESD	IEC 61000-4-2
Radiated	IEC 61000-4-3
EFT/Burst	IEC 61000-4-4
Surge	IEC 61000-4-5
Conducted	IEC 61000-4-6
Power frequency magnetic field	IEC 61000-4-8
Voltage dips and interruptions	IEC 61000-4-11
Harmonics	EN 61000-3-2, IEC 61000-3-2
Flicker	EN 61000-3-3, IEC 61000-3-3
Management	Web browser; SNMP Manager
Notes	Use only supported genuine HPE mini-GBICs with your switch.
Services	Refer to the Hewlett Packard Enterprise website at **hpe.com/networking/services** for details on the service-level descriptions and product numbers. For details about services, and response times in your area, please contact your local Hewlett Packard Enterprise sales office.

Standards and Protocols

(Applies to all products in series)

Device management	RFC 2819 RMON	Web UI
General protocols	IEEE 802.1D MAC Bridges IEEE 802.1p Priority IEEE 802.1Q VLANs IEEE 802.1s (MSTP) IEEE 802.1w Rapid Reconfiguration of Spanning Tree IEEE 802.3ad Link Aggregation Control Protocol (LACP) IEEE 802.3x Flow Control IEEE 802.3 Type 10BASE-T IEEE 802.3i 10BASE-T IEEE 802.3ab 1000BASE-T IEEE 802.3z 1000BASE-X	
MIBs	HC-ALARM-MIB SNMP-FRAMEWORK-MIB SNMP-NOTIFICATION-MIB SNMP-USER-BASED-SM-MIB SR-AGENT-INFO-MIB BRIDGE-MIB (IEEE 802.1Q) Q-BRIDGE-MIB (RFC 2674) LLDP-MIB (IEEE 802.3AB) LLDP-EXT-MED-MIB LAG-MIB (IEEE 802.3ad) RADIUS-ACC-CLIENT-MIB EtherLike-MIB IF-MIB (RFC 2863) RFC1213-MIB II Power Ethernet MIB (RFC3621)	
QoS/CoS	IEEE 802.1P (CoS)	
Security	IEEE 802.1X Port Based Network Access Control	

HPE OfficeConnect 1920S Switch Series accessories

Transceivers	$\boxed{\text{Aruba 1G SFP LC SX 500m MMF XCVR (J4858D)}}$ Aruba 1G SFP LC LX 10km SMF XCVR (J4859D) Aruba 1G SFP RJ45 T 100m Cat5e XCVR (J8177D) Aruba 100M SFP LC FX 2km MMF XCVR (J9054D)

Learn more at
hpe.com/networking

Hewlett Packard Enterprise

a00002986ENW, April 2018, Rev. 3

2.3. Wireless Access Point Meraki MR42

MR42

High performance 802.11ac Wave 2 wireless

The Cisco Meraki MR42 is a four radio, cloud-managed 3x3 MU-MIMO 802.11ac Wave 2 access point. Designed for next-generation deployments in offices, schools, hospitals, shops, and hotels, the MR42 offers performance, security, and simple management.

The MR42 provides a maximum 1.9 Gbps frame rate with concurrent 802.11ac Wave 2 and 802.11n 3x3:3 MIMO radios. A dedicated third radio provides real-time WIDS/WIPS with automated RF optimization. In addition, an integrated fourth radio delivers Bluetooth Low Energy (BLE) scanning and Beaconing functionality.

With a combination of cloud management, high perfomance hardware, multiple radios, and advanced software features, the MR42 makes an outstanding platform for the most demanding of uses today and tomorrow. These uses include high-density deployments and support for applications like voice and high-definition video.

MR42 and Meraki cloud management: a powerful combination

Management of the MR42 is handled through the Meraki cloud, enabling rapid deployment across multiple sites without the need for time-consuming training or costly certifications. Since the MR42 is self-configuring and managed over the web, it can be deployed at a remote location in a matter of minutes, even without on-site IT staff.

24/7 monitoring via the Meraki cloud delivers real-time alerts if the network encounters problems. Remote diagnostic tools enable immediate troubleshooting over the web, meaning multi site, distributed networks can be easily managed.

The MR42's firmware is automatically kept up to date via the cloud. New features, bug fixes, and enhancements are delivered seamlessly over the web. This means no manual software updates to download or missing security patches to worry about.

Product Highlights

- 3x3:3 MU-MIMO 802.11ac Wave 2
- 1.9 Gbps aggregate dual-band frame rate
- 24x7 real-time WIDS/WIPS and spectrum analytics via dedicated third radio
- Integrated Bluetooth Low Energy Beacon and scanning radio
- Enhanced transmit power and receive sensitivity
- Full-time WiFi location tracking via dedicated 3rd radio
- Integrated enterprise security and guest access
- Application-aware traffic shaping
- Optimized for voice and video
- Self-configuring, plug-and-play deployment
- Sleek, low-profile design blends into office environments

Features

Aggregate data rate of up to 1.9 Gbps

A 5 GHz 3x3:3 radio and a 2.4 GHz 3x3:3 radio offer a combined aggregate dual-band frame rate of 1.9 Gbps. Supports up to 1,300 Mbps in the 5 GHz band and 600 Mbps in the 2.4 GHz band. Technologies like transmit beamforming and enhanced receive sensitivity allow the MR42 to support a higher client density than typical enterprise-class access points, resulting in fewer APs for a given deployment.

Multi User Multiple Input Multiple Output (MU-MIMO)

With support for the 802.11ac Wave 2 standard, the MR42 offers MU-MIMO for efficient transmission to multiple clients. Especially suited for enviroments with numerous mobile devices, MU-MIMO enables multiple clients to receive data simultanously. This increases the total network perfomance and improves the end user experience.

Third radio delivers 24x7 wireless security and RF analytics

The MR42's sophisticated, dedicated dual-band radio scans the environment continuously, characterizing RF interference and containing wireless threats like rogue access points. No longer choose between wireless security, advanced RF analysis, and serving client data: a dedicated third radio means that all three occur in real-time, without any impact to client traffic or AP throughput.

Bluetooth Low Energy Beacon and scanning radio

An integrated fourth radio for Bluetooth Low Energy (BLE) provides seamless deployment of BLE Beacon functionality and effortless visibility of BLE devices. The MR42 enables the next generation of location-aware applications while futureproofing your deployment, making it ready for any new customer engagement strategies.

Automatic cloud-based RF optimization

The MR42's sophisticated, automated RF optimization means that there is no need for the dedicated hardware and RF expertise typically required to tune a wireless network. The RF analysis data collected by the dedicated third radio is continuously fed back to the Meraki cloud. This then automatically tunes the MR42's channel selection, transmit power, and client connection settings for optimal performance under even the most challenging RF conditions.

Integrated enterprise security and guest access

The MR42 features integrated, easy-to-use security technologies to provide secure connectivity for employees and guests alike. Advanced security features such as AES hardware-based encryption and WPA2-Enterprise authentication with 802.1X and Active Directory integration provide wire-like security while still being easy to configure. One-click guest isolation provides secure, Internet-only access for visitors. Our policy firewall (Identity Policy Manager) enables granular access control at the group or device level. PCI compliance reports check network settings against PCI requirements to simplify secure retail deployments.

Enterprise Mobility Management (EMM) & Mobile Device Management (MDM) integration

Meraki Systems Manager natively integrates with the MR42 to offer simple automatic security that is context aware. Rapidly deploy self-service MDM enrolment without installing additional equipment or dynamically tie firewall policies to client posture. End-to-end security has never been so easy.

Application-aware traffic shaping

The MR42 includes an integrated layer 7 packet inspection, classification, and control engine, enabling you to set QoS policies based on traffic type. Prioritize your mission critical applications, while setting limits on recreational traffic, e.g., peer-to-peer and video streaming. Importantly, controls can be implemented per network, per SSID, per user group, or per individual user.

Voice and video optmizations

Industry standard QoS features are easy to configure and come built in. Wireless Multi Media (WMM) access categories, 802.1p, and DSCP industry standards all ensure important applications get priorotized correctly, not only on the MR42, but on other steps in the traffic flow. Unscheduled Automatic Power Save Delivery (U-APSD) ensures minimal battery drain on wireless VoIP phones.

Low-profile, modern, user friendly design

Despite its extensive capabilities, the MR42 is packaged in a sleek, low-profile enclosure that blends seamlessly into any environment. This makes it ideal for modern offices, high end retail locations, and discrete deployments. Using human interface design principles, even the physical installation and mounting experience has been developed to eliminate error and simplify installation process.

Self-configuring, self-maintaining, always up-to-date

When plugged in, the MR42 automatically connects to the Meraki cloud, downloads its configuration, and joins the appropriate network. If new firmware is required, this is retrieved by the AP and updated automatically. This ensures the network is maintained with bug fixes, security updates, and new features managed for you.

Advanced analytics

Drill down into exceptional detail with highly granular traffic analytics: Understand how your network is used with access to numerous datasets. Extend your visibility to the physical world with journey tracking through location analytics. View vistor numbers, dwell time, repeat visit rates, and track trends. Fully customize your analysis with raw data available via simple APIs.

Specifications

Radios

2.4 GHz 802.11b/g/n client access radio

5 GHz 802.11a/n/ac client access radio

2.4 GHz & 5 GHz dual-band WIDS/WIPS, spectrum analysis, & location analytics radio

2.4 GHz Bluetooth radio with Bluetooth Low Energy (BLE) and Beacon support

Concurrent operation of all four radios

Max aggregate frame rate 1.9 Gbit/s

Supported frequency bands (country-specific restrictions apply):

2.412-2.484 GHz

5.150-5.250 GHz (UNII-1)

5.250-5.350 GHZ (UNII-2)

5.470-5.600, 5.660-5.725 GHz (UNII-2e)

5.725 -5.825 GHz (UNII-3)

Antenna

Integrated omni-directional antennas (5 dBi gain at 2.4 GHz, 5.5 dBi gain at 5 GHz)

Individual antenna elements for each radio

802.11ac Wave 2 and 802.11n Capabilities

3 x 3 multiple input, multiple output (MIMO) with three spatial streams

SU-MIMO and MU-MIMO support

Maximal ratio combining (MRC) & beamforming

20 and 40 MHz channels (802.11n), 20, 40, and 80 MHz channels (802.11ac)

Up to 256-QAM on both 2.4 GHz & 5 GHz

Packet aggregation

Power

Power over Ethernet: 37 - 57 V (802.3at required with functionality-restricted 802.3af mode supported)

Alternative 12 V DC input

Power consumption: 20W max (802.3at)

Power over Ethernet injector and DC adapter sold separately

LED Indicators

Multi color & multi function status indicator

Interfaces

1x 10/100/1000Base-T Ethernet (RJ45)

1x DC power connector (5.5 mm x 2.5 mm, center positive)

Mounting

All standard mounting hardware included

Desktop, ceiling, and wall mount capable

Ceiling tile rail (9/16, 15/16 or 1 ½" flush or recessed rails), assorted cable junction boxes

Bubble level on mounting cradle for accurate horizontal wall mounting

Physical Security

Two security screw options (included)

Kensington lock hard point

Concealed mount plate with anti-tamper cable bay

Environment

Operating temperature: 32 °F to 104 °F (0 °C to 40 °C)

Humidity: 5 to 95% non-condensing

Physical Dimensions

10.0" x 6.1" x 1.5" (253.4 mm x 155.8 mm x 37.1 mm), not including deskmount feet or mount plate

Weight: 25 oz (0.7kg)

Security

Integrated Layer 7 firewall with mobile device policy management

Real-time WIDS/WIPS with alerting and automatic rogue AP containment with Air Marshal

Flexible guest access with device isolation

VLAN tagging (802.1q) and tunneling with IPsec VPN

PCI compliance reporting

WEP, WPA, WPA2-PSK, WPA2-Enterprise with 802.1X

EAP-TLS, EAP-TTLS, EAP-MSCHAPv2, EAP-SIM

TKIP and AES encryption

Enterprise Mobility Management (EMM) & Mobile Device Management (MDM) integration

Quality of Service

Advanced Power Save (U-APSD)

WMM Access Categories with DSCP and 802.1p support

Layer 7 application traffic identification and shaping

Mobility

PMK and OKC credential support for fast Layer 2 roaming

Distributed or centralized layer 3 roaming

Analytics

Embedded location analytics reporting and device tracking

Global L7 traffic analytics reporting per network, per device, per application

Warranty

Lifetime hardware warranty with advanced replacement included

Ordering Information

MR42-HW: Meraki MR42 Cloud Managed 802.11ac AP

MA-PWR-30W-XX: Meraki AC Adapter for MR Series (XX = US, EU, UK or AU)

MA-INJ-4-XX: Cisco Meraki 802.3at Power over Ethernet Injector (XX = US, EU, UK or AU)

Note: Meraki access point license required.

Cisco Systems, Inc. | 500 Terry A. Francois Blvd, San Francisco, CA 94158 | (415) 432-1000 | sales@meraki.com

Compliance & Standards

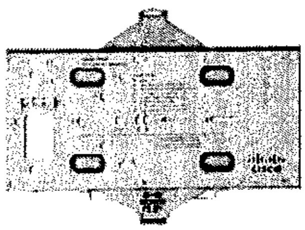
IEEE Standards
802.11b
802.11g
802.11a
802.11n
802.11ac
802.11h
802.11i
802.11e
802.11k
802.11r
802.11u

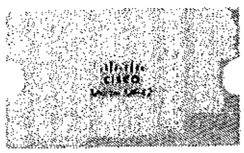
Safety Approvals
UL 60950-1
CAN/CSA-C22.2 No. 60950-1
IEC 60950-1
EN 60950-1
UL 2043 (Plenum Rating)

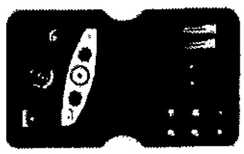
Radio Approvals
FCC Part 15C, 15E
RSS-247 (Canada)
EN 300 328, EN 301 893 (Europe)
AS/NZS 4268 (Australia/NZ)
NOM-121 (Mexico)
NCC LP0002 (Taiwan)
For additional country-specific regulatory information, please contact Meraki sales

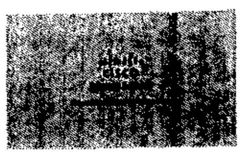
EMI Approvals (Class B)
FCC Part 15B
ICES-003 (Canada)
EN 301 489-1-17, EN 55032, EN 55024 (Europe)
CISPR 22 (Australia/NZ)
VCCI (Japan)

Exposure Approvals
FCC Part 2
RSS-102 (Canada)
EN 50385, EN 62311, EN 62479 (Europe)
AS/NZS 2772 (Australia/NZ)









RF Performance Table

2.4 GHz

Operating Band	Operation Mode	Data Rate	TX Power	RX Sensitivity
2.4 GHz	802.11b	1 Mb/s	21 dBm	-98 dBm
		2 Mb/s	21 dBm	-93.5 dBm
		5.5 Mb/s	21 dBm	-92 dBm
		11 Mb/s	21 dBm	-86 dBm
2.4 GHz	802.11g	6 Mb/s	21 dBm	-93 dBm
		9 Mb/s	21 dBm	-92.5 dBm
		12 Mb/s	20.5 dBm	-91 dBm
		18 Mb/s	20.5 dBm	-89 dBm
		24 Mb/s	19 dBm	-85 dBm
		36 Mb/s	19.5 dBm	-82.5 dBm
		48 Mb/s	18.5 dBm	-78 dBm
		54 Mb/s	18.5 dBm	-76 dBm
2.4 GHz	802.11n (HT20)	MCS0/8/16	21/24/25.7 dBm	-93/-96/-97.7 dBm
		MCS1/9/17	20.5/23.5/25.2 dBm	-89/-92/-93.7 dBm
		MCS2/10/18	20.5/23.5/25.2 dBm	-87/-90/-91.7 dBm
		MCS3/11/19	19/22/23.7 dBm	-83/-86/-87.7 dBm
		MCS4/12/20	19.5/22.5/24.2 dBm	-80/-83/-84.7 dBm
		MCS5/13/21	18.5/21.5/23.2 dBm	-76/-79/-80.7 dBm
		MCS6/14/22	18.5/21.5/23.2 dBm	-74/-77/-78.7 dBm
		MCS7/15/23	18/21/22.7 dBm	-73/-76/-77.7 dBm
2.4 GHz	802.11n (VHT20)	MCS0/0/0	21/24/25.7 dBm	-93/-96/-97.7 dBm
		MCS1/1/1	20.5/23.5/25.2 dBm	-89/-92/-93.7 dBm
		MCS2/2/2	20.5/23.5/25.2 dBm	-87/-90/-91.7 dBm
		MCS3/3/3	19/22/23.7 dBm	-83/-86/-87.7 dBm
		MCS4/4/4	19.5/22.5/24.2 dBm	-80/-83/-84.7 dBm
		MCS5/5/5	18.5/21.5/23.2 dBm	-76/-79/-80.7 dBm
		MCS6/6/6	18.5/21.5/23.2 dBm	-74/-77/-78.7 dBm
		MCS7/7/7	18/21/22.7 dBm	-73/-76/-77.7 dBm
		MCS8/8/8	17/xx/xx dBm	-73/xx/xx dBm
		MCS9/9/9	17/xx/xx dBm	-68/xx/x dBm

RF Performance Table

5 GHz

Operating Band	Operation Mode	Data Rate	TX Power	RX Sensitivity
5 GHz	802.11a	6 Mb/s	22 dBm	-92 dBm
		9 Mb/s	22 dBm	-91 dBm
		12 Mb/s	22 dBm	-90 dBm
		18 Mb/s	22 dBm	-88 dBm
		24 Mb/s	20 dBm	-84 dBm
		36 Mb/s	19 dBm	-81 dBm
		48 Mb/s	19 dBm	-76 dBm
		54 Mb/s	19 dBm	-74 dBm
5 GHz	802.11n (HT20)	MCS0/8/16	22/25/26.7 dBm	-92/-95/-96.7 dBm
		MCS1/9/17	22/25/26.7 dBm	-88/-91/-92.7 dBm
		MCS2/10/18	22/25/26.7 dBm	-86/-89/-90.7 dBm
		MCS3/11/19	22/23/24.7 dBm	-82/-85/-86.7 dBm
		MCS4/12/20	19/22/23.7 dBm	-79/-82/-83.7 dBm
		MCS5/13/21	19/22/23.7 dBm	-74/-77/-78.7 dBm
		MCS6/14/22	19/22/23.7 dBm	-73/-76/-77.7 dBm
		MCS7/15/23	19/22/23.7 dBm	-71/-74/-75.7 dBm
5 GHz	802.11n (HT40)	MCS0/8/16	2/25/26.7 dBm	-88/-91/-92.7 dBm
		MCS1/9/17	21.5/24.5/26.2 dBm	-85/-88/-89.7 dBm
		MCS2/10/18	20/23/24.7 dBm	-83/-86/-87.7 dBm
		MCS3/11/19	20/23/24.7 dBm	-79/-82/-83.7 dBm
		MCS4/12/20	19.5/22.5/24.2 dBm	-76/-79/-80.7 dBm
		MCS5/13/21	19.5/22.5/24.2 dBm	-72/-75/-76.7dBm
		MCS6/14/22	18.5/21.5/23.2 dBm	-70/-73/-74.7 dBm
		MCS7/15/23	18/21/22.7 dBm	-69/-72/-73.7 dBm

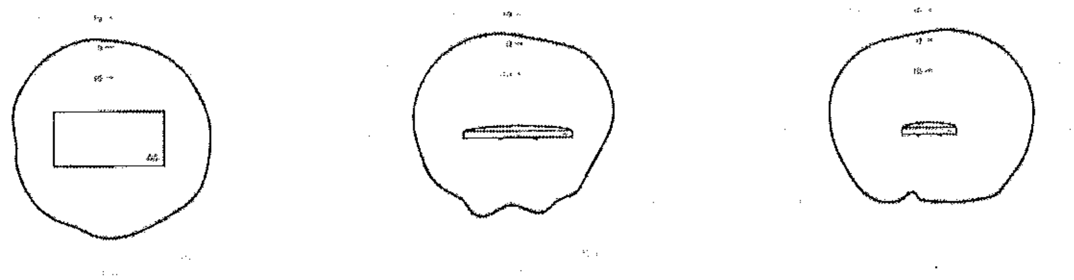
RF Performance Table

5 GHz

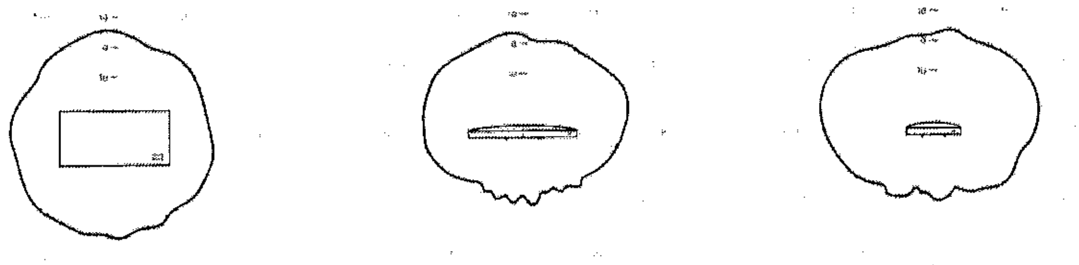
Operating Band	Operation Mode	Data Rate	TX Power	RX Sensitivity
5 GHz	802.11ac (HT20)	MCS0/0/0	22/25/26.7 dBm	-92/-95/-96.7 dBm
		MCS1/1/1	22/25/26.7 dBm	-88/-91/-92.7 dBm
		MCS2/2/2	22/25/26.7 dBm	-86/-89/-90.7 dBm
		MCS3/3/3	22/23/24.7 dBm	-82/-85/-86.7 dBm
		MCS4/4/4	19/22/23.7 dBm	-79/-82/-83.7 dBm
		MCS5/5/5	19/22/23.7 dBm	-74/-77/-78.7 dBm
		MCS6/6/6	19/22/23.7 dBm	-73/-76/-77.7 dBm
		MCS7/7/7	19/22/23.7 dBm	-71/-74/-75.7 dBm
		MCS8/8/8	18.5/21.5/23.2 dBm	-67/-70/-71.7 dBm
		MCS9/9/9	18.5/21.5/23.2 dBm	-63/-66/-67.7 dBm
5 GHz	802.11ac (HT40)	MCS0/0/0	22/25/26.7 dBm	-88/-91/-92.7 dBm
		MCS1/1/1	21.5/24.5/26.2 dBm	-85/-88/-89.7 dBm
		MCS2/2/2	20/23/24.7 dBm	-83/-86/-87.7 dBm
		MCS3/3/3	20/23/24.7 dBm	-79/-82/-83.7 dBm
		MCS4/4/4	19.5/22.5/24.2 dBm	-76/-79/-80.7 dBm
		MCS5/5/5	19.5/22.5/24.2 dBm	-72/-75/-76.7dBm
		MCS6/6/6	18.5/21.5/23.2 dBm	-70/-73/-74.7 dBm
		MCS7/7/7	18/21/22.7 dBm	-69/-72/-73.7 dBm
		MCS8/8/8	18/21/22.7 dBm	-67/-70/-71.7 dBm
		MCS9/9/9	18/21/22.7 dBm	-63/-66/-67.7 dBm
5 GHz	802.11ac (VHT80)	MCS0/0/0	22/25/26.7 dBm	-86/-89/-90.7 dBm
		MCS1/1/1	21.5/24.5/26.2 dBm	-82/-85/-86.7 dBm
		MCS2/2/2	21.5/24.5/26.2 dBm	-80/-83/-84.7 dBm
		MCS3/3/3	20.5/23.5/24.2 dBm	-76/-79/-80.7 dBm
		MCS4/4/4	20.5/23.5/24.2 dBm	-73/-76/-77.7 dBm
		MCS5/5/5	19.5/22.5/24.2 dBm	-69/-72/-73.7 dBm
		MCS6/6/6	19/22/23.7 dBm	-67/-70/-71.7 dBm
		MCS7/7/7	19/22/23.7 dBm	-66/-69/-70.7 dBm
		MCS8/8/8	18/21/22.7 dBm	-61/-64/-65.7 dBm
		MCS9/9/9	18/21/22.7 dBm	-59/-62/-63.7 dBm

Signal Coverage Patterns

Radiation Pattern for 2.4GHz Antennas



Radiation Pattern for 5GHz Antennas



2.4. Wireless Access Point Meraki MR74

MR74

General purpose industrial / outdoor 802.11ac Wave 2 wireless

The Cisco Meraki MR74 is a four-radio, cloud-managed 2x2 MIMO 802.11ac Wave 2 access point. Designed for general purpose, next-generation deployments in harsh outdoor locations and industrial indoor conditions, the MR74 offers performance, enterprise-grade security, and intuitive management.

The MR74 delivers a maximum 1.3 Gbps* aggregate frame rate with concurrent 2.4 GHz and 5 GHz radios. A dedicated third radio provides real-time WIDS/WIPS with automated RF optimization. A fourth radio delivers seamless Bluetooth Low Energy (BLE) scanning and Beaconing.

The combination of cloud management, 802.11ac, full-time RF environment scanning, and an integrated Bluetooth Low Energy radio delivers the high throughput, reliability, and flexibility required by the most demanding business applications like voice and high-definition streaming video, even in the most harsh outdoor environments.

MR74 and Meraki Cloud Management: A Powerful Combination

The MR74 is managed through the Meraki cloud, an intuitive browser-based interface that enables rapid deployment across multiple sites without the need for time-consuming training or costly certifications. Since the MR74 is self-configuring and managed over the web, it can be deployed at a remote location in a matter of minutes, even without on-site IT staff.

24x7 monitoring via the Meraki cloud delivers real-time alerts if the network encounters problems. Remote diagnostics tools enable immediate troubleshooting so that distributed networks can be managed with a minimum of hassle.

The MR74's firmware is always kept up to date from the cloud. New features, bug fixes, and enhancements are delivered seamlessly over the web, meaning no manual software updates to download or missing security patches to worry about.

Product Highlights

- Ideal for outdoor and industrial indoor environments
- 2x2:2 802.11ac, 1.3 Gbps aggregate dual-band data rate
- 24x7 real-time WIPS/WIDS and spectrum analytics via dedicated third radio
- Integrated Bluetooth Low Energy (BLE) Beaconing and scanning radio
- Forms point-to-point links with optional sector antennas
- Self-healing, zero-configuration mesh
- Integrated enterprise security and guest access
- Application-aware traffic shaping
- Self-configuring, plug-and-play deployment

*Refers to maximum over-the-air data frame rate capability of the radio chipset, and may exceed data rates allowed by IEEE Std 802.11ac-compliant operation.

Recommended Use Cases

Outdoor coverage for corporate campuses, educational institutions, metro Wi-Fi, and parks

- Weather-resistant Wi-Fi delivery in open spaces
- Monetizeable hotspots with built-in splash pages

Indoor coverage for industrial areas (e.g., warehouses, manufacturing facilities)

- Reliable coverage for scanner guns, security cameras, and POS devices
- High speed-access for iPads, tablets and laptops

Zero-touch point-to-point links

- Build a long-distance bridge between two networks
- Two MR74s can establish a long range link using high-gain antennas

Features

Dual-radio aggregate data rate of up to 1.3 Gbps*

An 867 Gbps 5 GHz 2x2:2 802.11ac radio and a 400 Mbps 2.4 ,Hz radio offer a combined aggregate dual-band throughput of 1.3 Gbps. Technologies like transmit beamforming and enhanced receive sensitivity allow the MR74 to support a higher client density than typical enterprise-class outdoor access points, resulting in fewer required APs for a given deployment. Band steering further enhances overall throughput, moving 5 GHz-capable clients to the 5 GHz radio and maximizing capacity in the 2.4 GHz range for older 802.11b/g clients.

Rugged industrial design

The MR74 is designed and tested for salt spray, vibration, extreme thermal conditions, shock and dust and is IP67-rated, making it ideal for extreme environments. Despite its rugged design, MR74 has a low profile and is easy to deploy.

Multi User Multiple Input Multiple Output (MU-MIMO)

The MR74 offers MU-MIMO (an 802.11ac Wave 2 standard) for efficient transmission to multiple clients. Especially suited for environments with numerous mobile devices, MU-MIMO enables multiple clients to receive data simultaneously. This increases the ɔtal network performance and improves the end user experience.

Bluetooth Low Energy Beaconing and scanning

An integrated fourth radio for Bluetooth Low Energy (BLE) provides seamless deployment of BLE Beaconing functionality and effortless visibility of BLE client devices within range of the network. The MR74 enables the next generation of location-aware applications while future-proofing your deployment for new user engagement strategies.

24x7 wireless security and RF analytics

The MR74's dedicated dual-band scanning and security radio continually assesses the environment, characterizing RF interference and automatically containing wireless threats like rogue access points. There's no need to choose between wireless security, advanced RF analysis, and serving client data: a dedicated third radio means that all three occur in real-time, without any impact to client traffic or AP throughput.

Integrated enterprise security and guest access

The MR74 features integrated, easy-to-use security technologies to provide secure connectivity for employees and guests alike. Advanced security features such as AES hardware-based encryption and WPA2-Enterprise authentication with 802.1X and Active Directory integration provide wire-like security while still being easy to configure. One-click guest isolation provides secure, Internet-only access for visitors.

Application-aware traffic shaping

The MR74 includes an integrated layer 7 packet inspection, classification, and control engine, enabling you to set QoS policies based on traffic type. Prioritize your mission critical applications, while setting limits on recreational traffic, e.g., peer-to-peer and video streaming. Importantly, controls can be implemented per network, per SSID, per user group, or per individual user.

Advanced Analytics

Drill down into the details of your network usage with highly granular traffic analytics. Extend your visibility into the physical world: View visitor numbers, dwell times, repeat visit rates, and compare trends. Fully customize your analysis with simple APIs.

High performance mesh

The MR74's advanced mesh technologies, like multi-channel routing protocols and multiple gateway support, make it possible to cover hard-to-wire areas and improve network resilience. In the event of a switch or cable failure, the MR74 will automatically revert to mesh mode.

Self-configuring, self-optimizing, self-healing

When plugged in, the MR74 automatically connects to the Meraki cloud, downloads its configuration, and joins the appropriate network. If new rmware is required, it is retreived by the AP and updated automatically. This ensures the network is maintained with bug fixes, security updates, and new features managed for you.

*Refers to maximum over-the-air data frame rate capability of the radio chipset, and may exceed data rates allowed by IEEE Std 802.11ac-compliant operation.

MR74 Tx / Rx Tables

2.4 GHz

Operating Band	Operating Mode	Data Rate	TX Power	RX Sensitivity
2.4 GHz	802.11b	1 Mb/s 2 Mb/s 5.5 Mb/s 11 Mb/s	20dBm 20dBm 20dBm 20dBm	-96dBm -93dBm -91dBm -89dBm
2.4 GHz	802.11g	6 Mb/s 9 Mb/s 12 Mb/s 18 Mb/s 24 Mb/s 36 Mb/s 48 Mb/s 54 Mb/s	20dBm 20dBm 20dBm 19dBm 19dBm 18dBm 18dBm 18dBm	-91dBm -90dBm -88dBm -87dBm -84dBm -81dBm -76dBm -75dBm
2.4 GHz	802.11n(HT20)	MCS0/8 MCS1/9 MCS2/10 MCS3/11 MCS4/12 MCS5/13 MCS6/14 MCS7/15	20/20 dBm 20/20 dBm 19/19 dBm 19/19 dBm 18/18 dBm 18/18 dBm 18/18 dBm 18/18 dBm	-91/91 dBm -88/-88 dBm -85/-85 dBm -82/-82 dBm -79/-79 dBm -75/-75 dBm -73/-73 dBm -70/-70 dBm
2.4 GHz	802.11n(HT40)	MCS0/8 MCS1/9 MCS2/10 MCS3/11 MCS4/12 MCS5/13 MCS6/14 MCS7/15	20/20 dBm 20/20 dBm 19/19 dBm 19/19 dBm 18/18 dBm 18/18 dBm 18/18 dBm 18/18 dBm	-89/-89 dBm -86/-86 dBm -84/-84 dBm -82/-82 dBm -77/-77 dBm -73/-73 dBm -71/-71 dBm -70/-70 dBm

MR74 Tx / Rx Tables
5 GHz

Operating Band	Operating Mode	Data Rate	TX Power	RX Sensitivity
5 GHz	802.11a	6 Mb/s 9 Mb/s 12 Mb/s 18 Mb/s 24 Mb/s 36 Mb/s 48 Mb/s 54 Mb/s	21dBm 21dBm 20dBm 20dBm 20dBm 20dBm 20dBm 20dBm	-90dBm -87dBm -86dBm -85dBm -84dBm -79dBm -74dBm -71dBm
5 GHz	802.11n(HT20)	MCS0/8 MCS1/9 MCS2/10 MCS3/11 MCS4/12 MCS5/13 MCS6/14 MCS7/15	21/21 dBm 21/21 dBm 20/20 dBm 20/20 dBm 20/20 dBm 20/20 dBm 20/20 dBm 19/19 dBm	-88/-88 dBm -85/-85 dBm -83/-83 dBm -79/-79 dBm -76/-76 dBm -72/-72 dBm -71/-71 dBm -69/-69 dBm
5 GHz	802.11n(VHT20)	MCS0/0 MCS1/1 MCS2/2 MCS3/3 MCS4/4 MCS5/5 MCS6/6 MCS7/7 MCS8/8	21/21 dBm 21/21 dBm 20/20 dBm 20/20 dBm 20/20 dBm 20/20 dBm 20/20 dBm 19/19 dBm 18/18 dBm	-88/-88 dBm -86/-86 dBm -83/-83 dBm -79/-79 dBm -77/-77 dBm -75/-75 dBm -72/-72 dBm -70/-70 dBm -67/-67 dBm
5 GHz	802.11n(HT40)	MCS0/8 MCS1/9 MCS2/10 MCS3/11 MCS4/12 MCS5/13 MCS6/14 MCS7/15	21/21 dBm 21/21 dBm 20/20 dBm 20/20 dBm 19/19 dBm 19/19 dBm 19/19 dBm 19/19 dBm	-85/-85 dBm -84/-87 dBm -84/-84 dBm -79/-79 dBm -77/-77 dBm -72/-72 dBm -70/-70 dBm -68/-68 dBm

Cisco Systems, Inc. | 500 Terry A. Francois Blvd, San Francisco, CA 94158 | (415) 432-1000 | sales@meraki.com

5 GHz	802.11n(VHT40)	MCS0/0 MCS1/1 MCS2/2 MCS3/3 MCS4/4 MCS5/5 MCS6/6 MCS7/7 MCS8/8 MCS9/9	21/21 dBm 21/21 dBm 20/20 dBm 20/20 dBm 19/19 dBm 19/19 dBm 19/19 dBm 19/19 dBm 18/18 dBm 17/17 dBm	-85/-85 dBm -82/-82 dBm -79/-79 dBm -77/-77 dBm -74/-74 dBm -70/-70 dBm -68/-68 dBm -67/-67 dBm -64/-64 dBm -63/-63 dBm
5 GHz	802.11ac(VHT80)	MCS0/0 MCS1/1 MCS2/2 MCS3/3 MCS4/4 MCS5/5 MCS6/6 MCS7/7 MCS8/8 MCS9/9	20/20 dBm 20/20 dBm 19/19 dBm 19/19 dBm 18/18 dBm 18/18 dBm 18/18 dBm 18/18 dBm 17/17 dBm 17/17 dBm	-83/-83 dBm -81/-81 dBm -79/-79 dBm -76/-76 dBm -73/-73 dBm -70/-70 dBm -67/-67 dBm -66/-66 dBm -62/-62 dBm -60/-60 dBm

Specifications

Radios
2.4 GHz 802.11b/g/n client access radio
5 GHz 802.11a/n/ac client access radio
2.4 GHz & 5 GHz WIDS/WIPS, spectrum analysis, and location analytics radio
2.4 GHz Bluetooth Low Energy (BLE) radio with Beacon and BLE scanning support
Concurrent operations of all four radios
Supported frequency bands (country-specific restrictions apply): 2.412-2.484 GHz 5.150-5.250 GHz (UNII-1) 5.250-5.350 GHZ (UNII-2) 5.470-5.600, 5.660-5.725 GHz (UNII-2e) 5.725 -5.825 GHz (UNII-3)

802.11ac and 802.11n Capabilities
× 2 multiple input, multiple output (MIMO) with two spatial streams
SU-MIMO and MU-MIMO support
Maximal ratio combining (MRC) & Beamforming
20 and 40 MHz channels (2.4Ghz), 20, 40, and 80 MHz channels (5Ghz)
Up to 256 QAM on both 2.4 GHz and 5 GHz bands
Packet aggregation

Power
Power over Ethernet: 37 - 57 V (802.3af compatible)
Power consumption: 11 W max (802.3af)
Power over Ethernet injector sold separately

Mounting
Mounts to walls and vertical poles
Mounting hardware included

Physical Security
:curity screw included

Environment
Operating temperature: -40 °F to 131 °F (-40 °C to 55 °C)
Humidity: 5 to 95% non-condensing
IP67 environmental rating

Physical Dimensions
10" x 6.2" x 2.6" (256 mm x 158 mm x 65 mm)
Weight: 2.4 lbs (1.09 kg)

Interfaces
1x 100/1000Base-T Ethernet (RJ45)
Four external N-type female antenna connectors

Security
Integrated layer 7 firewall with mobile device policy management
Real-time WIDS/WIPS with alerting and automatic rogue AP containment with Air Marshal
Flexible guest access with device isolation
VLAN tagging (802.1Q) and tunneling with IPSec VPN
PCI compliance reporting
WEP, WPA, WPA2-PSK, WPA2-Enterprise with 802.1X
EAP-TLS, EAP-TTLS, EAP-MSCHAPv2, EAP-SIM
TKIP and AES encryption
Enterprise Mobility Management (EMM) & Mobile Device Management (MDM) integration)

Quality of Service
Advanced Power Save (U-APSD)
WMM Access Categories with DSCP and 802.1p support
Layer 7 application traffic identification and shaping

Mobility
PMK, OKC, and 802.11r for fast Layer 2 roaming
Distributed or centralized layer 3 roaming

LED Indicators
1 power/booting/firmware upgrade status

Regulatory
RoHS
For additional country-specific regulatory information, please contact Meraki sales

Warranty
1 year hardware warranty with advanced replacement included

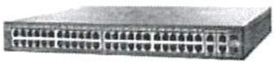
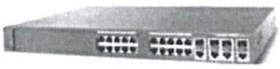
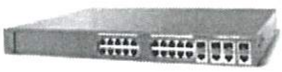
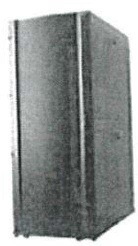
Ordering Information	
MR74-HW	Meraki MR74 Cloud Managed 802.11ac AP
MA-INJ-4-XX	Cisco Meraki 802.3at Power over Ethernet Injector (XX = US, EU, UK or AU)
MA-ANT-20	Meraki Dual-Band Omni Antennas
MA-ANT-21	Meraki 5 GHz Sector Antenna
MA-ANT-23	Meraki 2.4 GHz Sector Antenna
MA-ANT-27	Meraki Dual-Band Sector Antenna
MA-ANT-25	Meraki Dual-Band Patch Antenna

Note: Meraki Enterprise license required.

3. Compliance List

PROJECT NAME : JOTUN VIET NAM – GREEN FIELD PROJECT

DATA SHEET FOR TELECOMMUNICATION SYSTEM

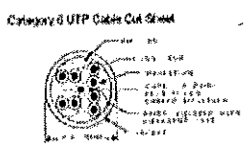
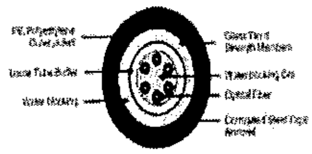
ITEM NO.	DESCRIPTION	OUTLINE DRAWING (FOR REFERENCE ONLY)	APPLICATION	Remark
1	**Switch 48-Port** - Suitable for power supply 230VAC, 50Hz - General colour as per manufacturer standard - Supported protocol: IPv6 and IPv4 - 48x10/100 Mbps Ports + 2 dual-purpose ports (10/100/1000 or SFP), LAN Base - Operating temperature: 32° to 104°F (0° to 40°C) - Operating humidity: 10% to 90%, relative, non-condensing - Suitable for installation to 19" Rack - Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
2	**Switch POE 48-Port** - Suitable for power supply 230VAC, 50Hz - General colour as per manufacturer standard - PoE Switch, Layer 2, Non- Stack Module - Supported protocol: IPv6 and IPv4 - 48x10/100 Mbps Ports + 2 dual-purpose ports (10/100/1000 or SFP), LAN Base - Operating temperature: 32° to 104°F (0° to 40°C) - Operating humidity: 10% to 90%, relative, non-condensing - Suitable for installation to 19" Rack - Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
3	**Switch 24-Port** - Suitable for power supply 230VAC, 50Hz - General colour as per manufacturer standard - Supported protocol: IPv6 and IPv4 - 24x10/100 Mbps Ports + 2 dual-purpose ports (10/100/1000 or SFP), LAN Base - Operating temperature: 32° to 104°F (0° to 40°C) - Operating humidity: 10% to 90%, relative, non-condensing - Suitable for installation to 19" Rack - Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
1	**Switch POE 24-Port** - Suitable for power supply 230VAC, 50Hz - General colour as per manufacturer standard - PoE Switch, Layer 2, Non- Stack Module - Supported protocol: IPv6 and IPv4 - 24x10/100 Mbps Ports POE+ 2 dual-purpose ports (10/100/1000 or SFP), LAN Base - Operating temperature: 32° to 104°F (0° to 40°C) - Operating humidity: 10% to 90%, relative, non-condensing - Suitable for installation to 19" Rack - Completed with Switch Rack Mount Bracket, power cable, data cable and other other accessories as required for a completely set.			
5	**Rack 42U (Free Standing) _ Series Server Cabinet** **- Rack 19" 42U, 800 x 2000 x 1000 (mm) (WxHxD)** - Material: Steel sheet. Frame 2mm thickness. Front door 1,5mm thickness. Side door: 1,2mm thickness. Rustproof, Electro Static painting. - Front door: 80% Flat Perforate steel 1,5mm with lock for security - Back door: Flat steel 1,5mm with two small half door - Incoming/outgoing cable shall be located at rear bottom and top of Rack - Static loading: 1000kg - Colour: Black - Standard: ANSI/EIA 310-D, Type A, 1U = 44.45 mm (1.75") DIN 41494 ,BS5954 Part 2,IEC 60297-1, IEC 60297-2 - Completed with ventilation fans, horizontal cable management, power distribution unit and other accessories as required for a completely set. - Completed with fix Shelf Depth 850, Black - Name plate for LAN Rack shall be stainless steel			

Comply

Comply

PROJECT NAME : JOTUN VIET NAM – GREEN FIELD PROJECT

DATA SHEET FOR TELECOMMUNICATION SYSTEM

ITEM NO.	DESCRIPTION	OUTLINE DRAWING (FOR REFERENCE ONLY)	APPLICATION	Remark
13	**Wireless Access Point Meraki MR74** **+Complied with :** -Meraki Dual-band Omni Antennas (4 poles antennas) -Meraki MR74 Cloud Managed AP -Meraki MR Enterprise License, 5YR		Warehouse Area	
14	**Wireless Access Point Meraki MR33** **+Complied with :** -Meraki MR33 Cloud Managed AP -Meraki Dual-band Omni Antennas -Meraki MR Enterprise License, 5YR		Office Area	
15	**Cat 6 Modular Plug RJ45** - Modular Plug Cat 6, 8 Pins, 23-24AWG, for Solid Cable, 6-7mm - Completed with Modular Plug Boot RJ-45, Blue - General colour as per manufacturer standard			
16	**Cat 6 Patch Cord** - 4 Pair stranded UTP cable - CM Jacket - UTP Cat 6 SL Series Patch Cord 24AWG 10FT (3M) - Colour: Blue			
17	**Fiber Optic Cable Patch Cord** - Duplex SC to Duplex SC, OM3, 10FT (3M). - Colour: Aqua			
18	**Category 6 UTP Cable** - Cat 6 unsheild twisted pairs cable - Conductor: 23 AWG Solid Copper - Insulation: Polyethylene (PE) - Jacket: LSHF 4 - Fire Rating: CM - Standard: TIA/EIA-568-B.2-1 Category 6 - Outer jacket colour: Blue			Comply
9	**Outdoor Fiber Optic Cable** - Multi-mode Fiber Optic (6C) Outdoor, OM3, 50/125 micro metre - Completed with corrugated steel tape armored and louse tube buffer - Outer jacket colour: Black			

TEMPLATE

Date:

PRODUCT WARRANTY CERTIFICATE

The Contractor: SHINRYO VIETNAM CORPORATION

The Employer: JOTUN PAINTS (VIETNAM) CO., LTD

Project name: JOTUN VIETNAM GREENFIELD PROJECT

Location: Lot 3, Hiep Phuoc Industrial Park, Nha Be Commune, Ho Chi Minh City

Scope of works: SUPPLYING OF TELECOMMUNICATION SYSTEM AS PER PURCHASE ORDER NO.6001403200-038

We hereby warrant that the product supplied under the above Purchase Order is new and in accordance with the approved technical specifications. Enclosed attachment is detailed.

We further warrant the above - mentioned goods have no defected, and have been passed the factory test that may develop under normal use of supplied goods in the conditions prevailing in Vietnam.

Warranty period for the works shall be 24 months from 30 April 2021 to 30 April 2023 and free from any defects under normal operation.

The defective works, if founds any, shall be remedy free of charge during the warranty period.

The warrantee does not cover any defect caused by misuse or abuse of the works, damage caused by natural disasters, damage caused by unauthorized modification, tear and wears or any such act of negligence.

Best regards,

<SUPPLIER NAME>

<Signature and full name>

innotel
maximizing your access

THUAN PHONG INNOVATIVE TELECOMMUNICATION CO., LTD.
Website: http://www.innotel.com.vn Email: sales@innotel.com.vn
Tel: (84 8) 38621119. Fax: (84 8) 38630366.
Duy Tan Plaza, Suite 92, 3/2 Street, Dist 10., HCMC, Vietnam

BẢNG CHÀO GIÁ

Kính gửi:	**SHINRYO VIETNAM CORPORATION**						
Người nhận:	BỘ PHẬN TENDER						
TEL:							
FAX NUMBER:							
MOB:							
Người gửi:	Huỳnh Trọng Hiếu						
Ngày :	13-Thg6-20				Page:	1
Chủ đề	**TBM - JOTUN**				Qno:	

STT	Mã hàng	Tên hàng	Nhãn hiệu	Xuất xứ	Bảo hành	SL	Đơn giá(VNĐ)	Thành tiền(VNĐ)
1	**Switch**							-
1,1	J9772A	Aruba 2530 48G PoE+ Switch	HPE	Asia	Đến 30-4-2023	9	37.000.000	333.000.000
1,2	JL385A	HPE 1920S 24G 2SFP PoE+ 370W Switch	HPE	Asia	Đến 30-4-2023	8	17.800.000	142.400.000
1,3	J4858D	Aruba 1G SFP LC SX 500m MMF Transceiver	HPE	Asia	Đến 30-4-2023	34	3.980.000	135.320.000
2	**Access Point**						-	-
2,1	MR42-HW	Meraki MR42 Cloud Managed AP	Cisco-Meraki	Asia	5 năm	18	14.500.000	261.000.000
2,2	MR74-HW	Meraki MR74 Cloud Managed AP	Cisco-Meraki	Asia	5 năm	51	19.890.000	1.014.390.000
	MA-ANT-20	Meraki Dual-band Omni Antennas	Cisco-Meraki	Asia	5 năm	102	2.880.000	293.760.000
2,3	LIC-ENT-5YR	Meraki MR Enterprise License, 5YR	Cisco-Meraki	Asia	5 năm	69	6.480.000	447.120.000
3	**Dịch vụ cấu hình**		Innotel	vietnam		1	67.000.000	67.000.000
							TỔNG VND	2.693.990.000
							VAT (10%)	269.399.000
							THÀNH TIỀN	2.963.389.000

CÁC ĐIỀU KHOẢN THƯƠNG MẠI

1. Giao hàng tại công trình
2. Thời gian giao hàng: 6-8 tuần, bảo hành theo tiêu chuẩn nhà sản xuất
3. Thanh toán 90% khi giao hàng trong vòng 15 ngày khi nhận hàng tại công trình.
   Đợt 2 thanh toán tiếp 10% giá trị hợp đồng sau khi T&C và bàn giao thư bảo hành.

CÔNG TY TRÁCH NHIỆM HỮU HẠN VIỄN THÔNG SÁNG TẠO THUẬN PHONG

Huỳnh Thanh Hải
Giám đốc

Xác nhận đặt hàng